MATH INQUIRERS HOLIDAY PROGRAM

Math Inquirers Level 2 Tutor Guide

Holiday program for Level 2

Notice it. Test it. Explain why.

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Program guide for tutors: schedule, lesson routine, ten minute-by-minute lesson scripts with answer keys, a progress tracker and a parent summary. Math Inquirers is online first: every lesson runs live on video; the face-to-face notes are kept for partner centres.

The program at a glance

Level	Children whose school year covers Primary 2 topics (ages 8 to 9), going into P3, who are ready for more thinking than standard practice
Length	10 lessons of 90 minutes, two a week over the 5 weeks of the holidays
Group size	Up to 8 children per tutor, live online. Pilot with 6. Add a second tutor or helper above 8.
Focus	Thinking skills: making a list in order, guess and check, drawing a model, working backwards, finding patterns, reasoning and explaining. Not speed drills.
The arc of every lesson	Start concrete with objects, record in a table, spot the pattern, state the rule, use it (often backwards), then explain why.
What children take home	A student workbook with an inquiry journal, 10 worksheets and a certificate
Before tutors start	Every tutor signs the tutor confidentiality agreement before receiving any materials.

Schedule

Week	Lesson	Title	Topic	Big idea
Week 1a	1	**Digit Detectives**	Numbers to 1000	Place value
Week 1b	2	**Missing Digit Mysteries**	Addition and Subtraction within 1000	Working backwards

Week 2a	3	**Legs, Wheels and Arrays**	Multiplication and Division	Draw it (arrays)
Week 2b	4	**Pattern Machines**	Patterns with Multiplication	Look for a pattern
Week 3a	5	**Mystery Number Machines**	Working Backwards	Inverse operations
Week 3b	6	**Market Day**	Money	Systematic listing (make a list in order)
Week 4a	7	**Fair Shares**	Fractions	Checking for equal parts
Week 4b	8	**Shape Builders**	2D and 3D Shapes	Spatial reasoning
Week 5a	9	**Measure Masters**	Length, Mass, Volume and Time	Reasoning with equal amounts (swapping)
Week 5b	10	**Challenge Day**	Mixed Puzzles	Choosing a strategy

Suggested pairing: two lessons a week, for example Tuesday and Thursday, or Saturday and Sunday. Keep the same time slot all five weeks.

The 90-minute lesson routine

Every lesson uses the same routine so children know what to expect and tutors can prepare quickly.

Minutes	Part	What happens	Face to face	Online
0–8	**Warm-up**	Welcome (0–1). Sneak-peek puzzles (1–4): two puzzles on ideas not learnt yet, answers shown only to the tutor, nobody marked. A 2-minute game (4–6). Quick questions (6–8). Everyone joins in.	Sit in a circle or horseshoe.	Cameras on. Answers on own paper shown to the tutor on camera, by thumbs, or kept to share if the child wants. Nothing is sent by private message. No whiteboards raised together.
8–20	**Explore**	Hands-on in pairs. Objects first, no writing yet.	Materials box on each table.	Each child has their own materials at home. Pairs go to breakout rooms.
20–32	**Mini-lesson**	Tutor builds the table on the board, names the rule and asks what stays the same and what changes.	Whiteboard.	Shared slide and annotation tool.
32–36	**Brain break**	Stand, stretch, drink water (32–34). Optional Story break (34–36): a 2-minute story about a mathematician, with a try-it question. Skip it whenever time is short.	A short movement game.	Stand up on camera. Count the stretches together.
36–72	**Worksheet**	Page 1 in pairs (36–50), check together (50–52), short teaching (52–57), page 2 alone (57–69), check together (69–72). Tutor asks “how do you know?” rather than giving answers.	Printed workbook, tutor circulates.	Printed workbook. Child holds the page up to the camera. Tutor checks one or two answers each.
72–77	**Talk time**	Volunteers share one answer and explain it. Compare methods. Any child may pass. No cold-calling by name, no public tallies or rankings.	The tutor invites volunteers first.	The tutor invites volunteers; a child unmutes when invited. The whole group is invited to share (a child may pass); quiet children are welcome in the open session with a parent nearby, and are never singled out.

77–80	**Investigation share**	Three minutes on the Investigation from the last core page (its own slide). The tutor asks the whole group for one way someone tried it; several answers are right. Pass is always fine. If nobody has started, pairs try the first step for one minute and the rest is take-home.	The tutor invites volunteers first.	The tutor invites volunteers; a child unmutes when invited.
80–90	**Exit ticket, Looking back and stamp**	Exit ticket (80–87, one question on a slip of paper). Looking back (87–89, a small box on the last core page, a pass is always allowed). Closing and stamps (89–90).	Paper slip, tutor stamps.	Child writes the answer and shows the tutor only. Tutor stamps the next time or posts it.

Teaching habits that matter

- Ask “How do you know?” after answers, right or wrong, but invite rather than single out: “Would anyone like to say how they started?” “I am not sure yet” is always a fine answer.
- Wait 10 seconds before helping. Let children struggle a little.
- Praise the thinking (“You checked it two ways”), not just the answer.
- Let children disagree with each other and settle it with objects or a drawing.
- If a child would like more, offer Think harder, Go further or the Investigation as a choice. Never name who is on which page, and never use extra drill.

Preparing the materials

- Send the workbook PDF to parents one week before. Ask them to print it single-sided and the cut-out pages on card if they can.
- Cut-outs used: place-value cards (Lessons 1, 2, 10), digit cards (Lessons 1 and 10), play coins and notes (Lesson 6), fraction strips (Lesson 7), cube nets and paper cube strips (Lesson 8), paper clock (Lesson 9).
- Keep the answer key with the tutor only. Check answers together on screen.
- Use the shared screen or a digital whiteboard for the mini-lesson table.

Setting up online

- Send parents the workbook and materials list one week before. Needed: the printed workbook and cut-outs, 20 counters or small objects, scissors, a split pin or short pencil for the clock, paper and a pencil.
- Use one approved platform with breakout rooms and screen share. Admit only registered children and never share the link publicly.
- Ask the parent to sit nearby but not give answers. The camera should show the child’s page and hands.
- At the end of each lesson a child may show their page to the tutor on camera, or keep it to share if they want. No photos of children’s work are sent to or from the tutor.

- Do not record lessons (this is in the tutor agreement). Add a second tutor or helper above 8 children.

Materials master list

Lesson	Title	Face to face	Online set-up
1	Digit Detectives	Digit cards 0 to 9 (one set per child), place-value mat (hundreds, tens, ones), base-ten blocks or paper strips and squares, mini whiteboards, a 1000 chart page (rows of 10).	Children cut out paper digit cards before the lesson and move them on camera; the tutor writes on the slide 1000-chart pieces.
2	Missing Digit Mysteries	Digit cards 0 to 9, base-ten blocks or place-value counters, mini whiteboards, a paper magic triangle with 6 circles and number tiles 110 to 160.	Children write column sums on whiteboards or paper and show your tutor on camera or keep it to share if you want; number tiles are cut from paper before the lesson. The tutor writes on the slide sums.
3	Legs, Wheels and Arrays	About 30 counters per pair, mini whiteboards, squared paper. Optional: picture cards of hens and goats, bicycles and tricycles.	Each child has 30 small objects (buttons, dried beans or Lego bricks) and a whiteboard or paper. The tutor shares the slides; children show their tutor on camera or keep it to share if you want (arrays and tables).
4	Pattern Machines	Linking cubes in 3 colours (about 20 per pair), mini whiteboards, matchsticks or lolly sticks (about 25 per pair), counters.	Children use coloured pencils and squared paper (or Lego bricks) for the cube train, and draw sticks for the growing pattern. The tutor shares the slides and the children show their tutor their tables on camera or keep them to share if they want.
5	Mystery Number Machines	Mini whiteboards, counters, paper cups, number machine cards (× 2, × 3, × 4, × 5, × 10 and + or − cards from 1 to 20), paper strips for bars.	Machine cards shown on the slide; children write on whiteboards or paper and show their tutor on camera or keep it to share if they want. For bars, children draw on paper or fold paper strips into equal parts.

6	Market Day	Play money: coins 5¢, 10¢, 20¢, 50¢, $1 and notes $2, $5, $10, $50 (about 6 of each per pair), price tags, mini whiteboards.	Print the cut-out coins and notes, or use real money at home. No play money? Draw each coin and note on paper. The shop and the toy stall are on the slides; children show their money to the camera.
7	Fair Shares	Paper strips of the same length (4 per child), paper squares, scissors, crayons, mini whiteboards.	Each child folds paper strips on camera; the tutor shows fraction bars on the slides and children colour their own bars.
8	Shape Builders	Paper circles (2 per child), scissors, about 12 linking cubes per child (or sugar cubes), the cube-net cut-out page (nets A to F, near the back of the workbook), mini whiteboards.	Children cut circles and build stacks on camera; the six nets A to F are on a cut-out page near the back of the workbook (print and cut them before the lesson), or cut a net from an empty box. No linking cubes at home? Use sugar cubes, dice or small boxes, or draw each stack layer by layer.
9	Measure Masters	A coat hanger, two paper cups, string, a glue stick, a spoon and about 20 20¢ coins (a home-made balance; backup: a ruler balanced on a pencil); for the jug puzzles two cups or bottles of different size (about 3 and 5 units), a jug of dry rice or sand (or water), and tape marking 3 and 5 units; a cardboard clock with hands; mini whiteboards.	Children build the coat-hanger balance before the lesson and hold it up to the camera. No coat-hanger balance at home? The tutor balances a ruler on a pencil on camera and pairs share one view. The tutor shares the slide balances, jugs and clocks and annotates on screen. Each child needs a cardboard or paper-plate clock. For the pouring puzzles use dry rice or sand rather than water when online, and let children act out each pour with the two containers.
10	Challenge Day	Four station cards (one slide each in the deck, and printed in the tutor guide), play coins, digit cards 0 to 9, paper strips for fractions, mini whiteboards, certificates, stamps.	Each breakout pair gets one station card (the tutor pastes its text into the breakout room chat), works for 3 minutes, then swaps with another pair for a second card, and everyone reports back to the group. Each child needs digit cards (cut from paper), a pencil and a mini whiteboard.

Level 2 stories: mathematicians

Each lesson has an optional 2-minute Story break at the brain break (minutes 34 to 36): one slide titled “Story break: <name>”, a small Story corner box on the “Before you go” page of the workbook, and a try-it question. The answers to the try-it questions are in each lesson’s answer key below, never on the child page. Where a story is a legend or told long after the event, the slide says so: say it out loud too.

Lesson	Person	Country	Years	Link to the lesson
1	**Aryabhata**	India	476 - about 550 (sources differ)	Aryabhata wrote that each place in a number is worth ten times the place before it, which is the idea that tells you what each digit is worth in a 3-digit number.
2	**Carl Friedrich Gauss**	Germany	1777-1855	Gauss's pairing trick shows that adding can be quick when you spot numbers that make a friendly total, which helps when you add and balance number sentences today.
3	**Shakuntala Devi**	India	1929-2013	Shakuntala Devi multiplied numbers quickly in her head, and you build the same skill by learning your times tables and arrays today.
4	**Hertha Ayrton**	Britain (born in Portsea, England)	1854-1923	Sand ripples make a pattern that repeats in a regular way, and you find far places in repeating patterns by counting in groups.
5	**Mary Somerville**	Scotland (and later Italy)	1780-1872	The X and Y in her magazine stood for mystery numbers, just like the unknown numbers you find by working backwards today.
6	**Luca Pacioli**	Italy	about 1447 - 1517 (sources differ)	Pacioli taught merchants to write down what they spent and what they had left, just like keeping track of money and change on Market Day.
7	**Ahmes the scribe**	Egypt	about 1550 BC (date on the papyrus copy; sources differ)	Ahmes's roll shows how people long ago wrote and used fractions, the same fractions you share and compare today.
8	**Alicia Boole Stott**	Ireland (born in Cork; later lived in England)	1860-1940	Alicia made solid shapes by joining flat pieces of card, the same idea as folding a net into a 3D shape.

9	**Thales of Miletus**	Miletus, a Greek city (now in Turkey)	about 624 - about 547 BC	The story is about measuring a length by using a shadow and choosing the right time of day, which links to measuring length and telling the time.
10	**Paul Erdos**	Hungary	1913-1996	Erdos shows that tricky problems are easier and more fun when you share ideas with a partner, which is how Challenge Day works.

Rules for the Story break

- It is always optional. Skip it, without comment, if you are behind time, the group is tired or the break needs all four minutes. Nothing later depends on it.
- Read the story aloud (about 30 seconds), then show the try-it question. Ask “Would anyone like to try this one?” Take volunteers only, never name a child, nobody is marked, and passing is always fine.
- Children may show an answer on their own paper to you only, or tell a partner. No public tallies or rankings. The same safeguarding rules apply as in the rest of the lesson.
- Keep the stories true: say “a story often told” or “historians are not sure” where the slide says so (Gauss, Mary Somerville, Thales; dates for Aryabhata, Pacioli and Ahmes differ between sources).
- The try-it questions use only Primary 2 ideas. The stories mention some big numbers (for example 5050 or a 13-digit number) as history, not as questions for the children.

How to use the scripts

Every lesson below is a minute-by-minute script. The slides match it: one slide or two for each part. You do not have to say the words exactly, but keep the questions, because the questions are what makes children think.

SAY	Words to say aloud, in plain language for 8- and 9-year-olds. Said in your own voice, not read from the page.
DO	What you do: set up, hand out, show a slide, or move around the room.
ASK / EXPECT	Questions to put to the group, with the answer you are listening for. Let children answer before you explain.
IF STUCK	A ladder of help. Give hint 1, wait ten seconds, then give hint 2. Do not give the answer.

Five habits that run through every script

- Predict first. Before children test anything, ask for a guess and write it down. A wrong guess is useful because we can test it.
- Wait ten seconds after every question. Count silently. Most children answer in the last four.
- Ask “How do you know?” after right answers too (to volunteers), so that children do not read your face for the answer.
- Praise the thinking, not the answer: “You checked it two ways” beats “Well done”.
- When an answer is wrong, say “Let us test that,” and use objects, not “No”.

Answering a wrong answer

If...	Then...
A child gives a wrong answer	“Thank you. Let us test it with the counters.” Then build it together.
A child gives a right answer but cannot explain	“Show me with your counters.” Then: “Tell your partner how you know.”
A child says “I do not know”	“What do you notice? What could you try first?” Offer hint 1.
Two children disagree	“Both of you, convince the group using objects.” Let the objects decide.

Lesson 1: Digit Detectives

Week 1 (first lesson) • Numbers to 1000 • Workbook: Lesson 1

Objective	Read and build 3-digit numbers by place value (also with more than 9 tens), make every number from a set of digit cards by listing in order, find the biggest and smallest, and use patterns on a 1000 chart to count on and back in 1s, 10s and 100s.
Thinking skills	Place value, making a list in order, looking for a pattern, reasoning about zero
Materials	Digit cards 0 to 9 (one set per child), place-value mat (hundreds, tens, ones), base-ten blocks or paper strips and squares, mini whiteboards, a 1000 chart page (rows of 10).
Online set-up	Children cut out paper digit cards before the lesson and move them on camera; the tutor writes on the slide 1000-chart pieces.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0–1 min Opening

SAY *"Welcome, Digit Detectives! Every 3-digit number is made of just three digits, but WHERE a digit sits changes everything."*

"Today we find every number hiding in a set of digit cards, and we make sure we do not miss any. Notice it. Test it. Explain why."

DO Greet each child by name. Show the lesson title slide. Read "Today we will..." aloud.

1–4 min Sneak-peek puzzles (3 minutes)

SAY *"These are sneak-peek puzzles. They are ideas we have not learnt yet, so guessing is welcome and so is saying "I do not know". Nobody is marked and nothing is read out. Start with question 1: use objects, a drawing or counting. Then try question 2. Show your answer on your own paper to your tutor on camera, or keep it to share if you want. If you would like help, the tutor will invite the whole group to ask in the open session, and a parent can stay nearby."*

DO Show the Sneak-peek puzzles slide at minute 1 (3 minutes: about 1 minute for question 1, then 2 minutes for question 2). Say out loud: these are ideas we have not learnt yet; guessing and "I do not know" are welcome; nobody is marked.

Answers stay private. Each child holds their own paper to the camera for you only (look, do not read out), or writes it on their own paper and shows only you on camera. Never raise whiteboards together and never show a board in front of the group. Never ask a named child "how do you know?" Ask for volunteers: "Would anyone like to tell us how they started?"

Question 1 is an access question: every child can try it with objects, a drawing or counting, so everyone has a first win. Question 2 is the thinking question. Do not mark either one. Click to show the answers at the end as "one way to think about it", without saying who had what.

Use what you see only to plan. Invite the WHOLE group at once, never one child: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Never announce who is on which page and never name a child in front of the group. Children choose, and may move either way at any time. Pairs are not split unless both children agree. Everybody joins the explore, the mini-lesson, the check-together slides and talk time.

Sneak-peek puzzle	Answer and what to listen for
Question 1 (access: objects, drawing or counting): Mei has 3 hundreds, 2 tens and 5 ones. Draw them if you like. What number is that?	325. Listen for: Hundreds first, then tens, then ones. A drawing of 3 flats, 2 rods and 5 units is welcome. This is a first win for everyone.
Question 2 (thinking): Mei has digit cards 4, 0 and 7. She uses each card once to make a 3-digit number. How many different numbers can she make? Write them.	4 numbers: 407, 470, 704, 740. Listen for: 0 cannot be the hundreds digit, so only 4 and 7 can start. Listing in order: 4 starts 2 numbers, 7 starts 2 numbers. A child who says 6 is using 3 x 2 x 1 without the zero rule: a good idea, not a mistake to correct yet.

DO The neutral line “After question 3 you may choose: carry on, or try Think harder.” is printed on page 1 of the workbook, next to the lesson title. Nobody is labelled and nothing is announced.

4–8 min Warm-up

DO Place It Right. The tutor turns over digit cards one at a time. Write each one in the hundreds, tens or ones box straight away. No moving! Biggest number wins.

SAY *“Draw three boxes on your whiteboard: hundreds, tens, ones. I will turn over one digit card at a time. As soon as you see it, write it in a box. No moving it later!”*

“Try to make the BIGGEST number. Where will you put a 9? Where will you put a 1? Why?”

DO Keep the game to two minutes (4 to 6). Then the quick questions (6 to 8): ask questions 1, 3 and 5; questions 2 and 4 are spare. Children answer by thumbs, a whisper to a partner. Do not name a child.

Ask	Expect
4 hundreds, 6 tens, 2 ones = ?	462
What is the 8 worth in 384?	80
10 more than 296?	306
100 less than 405?	305
The biggest 3-digit number?	999

8–20 min Explore (in pairs, with objects)

SAY *“You have the cards 3, 5 and 8. How many different 3-digit numbers do you think you can make with them? Write your guess in the corner of your whiteboard.”*

DO 1. Take the digit cards 3, 5 and 8.

2. Make a 3-digit number using each card once. Write it down.

3. Make as many different numbers as you can.

4. Circle the biggest and the smallest. Have you got them all?

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK Stuck: put the 3 in the hundreds place first. What can the other two cards do?

Not sure if you have them all: "How many numbers start with 3? With 5? With 8?"

Finished early: add a 0 card and make numbers with 0, 3 and 5. Why are there fewer?

DO Go further (a second what-if, so nobody waits): show the Explore: go further slide or post it in the chat when pairs go to their rooms. Any child may try it any time, as a choice. Never name who does. Question: add a fourth card, 2. Use any THREE of the cards 2, 3, 5 and 8. How many different 3-digit numbers can you make? Answer: 24: 4 choices of which card to leave out, and each set of 3 cards makes 6 numbers (4 x 6 = 24).

DO At about minute 14, say "Pencils down" and bring the group back together. Take the three questions below, one at a time (about 6 minutes).

Ask	Expect
How many numbers did you make?	6: 358, 385, 538, 583, 835, 853.
How do you know you have them all?	List in order: each card starts 2 numbers, and 3 × 2 = 6.
Which is the biggest? The smallest?	853 and 358: put the biggest digit first for the biggest.

20–32 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"5 hundreds, 12 tens and 4 ones. 12 tens is 120, so the number is 500 + 120 + 4 = 624. I can also see it as 6 hundreds, 2 tens, 4 ones, because 10 of the tens make another hundred."*

"Cards 1, 6 and 9. I keep the hundreds digit the same and swap the other two. 1 first: 169, 196. 6 first: 619, 691. 9 first: 916, 961. Each card starts 2 numbers, and 3 cards make 6 numbers. That is how I know I have them all."

"The biggest number has the biggest digit in the hundreds place: 961. The smallest has the smallest digit in the hundreds place: 169. But zero can never go first: 036 would be just 36 (thirty-six), which has only 2 digits, so it is not a 3-digit number."

Worked example on the slides	Answer
Place value: more than 9 tens. 5 hundreds, 12 tens and 4 ones. 12 tens = 120. 500 + 120 + 4 = ?	624. Ten of the tens make another hundred.

Every number from 1, 6, 9. Fix the hundreds digit, then swap the other two. 1 first: 169, 196. 6 first: 619, 691. 9 first: 916, 961. How many? Which is biggest?	6 numbers. Biggest 961, smallest 169.

SAY *"The key idea today is: list in order. Fix the hundreds digit, then swap the others. Each card starts the same number of numbers. Say it with me."*

DO Show the key idea slide at about minutes 31 to 32. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Writing 6 hundreds, 15 tens and 2 ones as 6152 instead of regrouping 15 tens as 1 hundred and 5 tens (752).
- Listing digit-card numbers at random, so one is missed or written twice, or counting 059 as a 3-digit number.
- Counting back in 10s across a hundred: 512, 502, 592 instead of 512, 502, 492.

32–36 min Brain break

DO Swap the digits. Each child puts three digit cards in a row (hundreds, tens, ones). The tutor calls "swap the hundreds and the tens" or "swap the tens and the ones". Show your new number on camera. Is it bigger or smaller?

Water. Toilet break for anyone who needs it. Keep it to four minutes.

DO Story break (optional, 2 minutes): do the movement game for 2 minutes (32 to 34), then show the "Story break: Aryabhata" slide (34 to 36). Read the story aloud (about 30 seconds): "Aryabhata lived in India about 1,500 years ago. In the year 499 he wrote a short book of maths and astronomy, all in verses. One rule in his book says that each place in a number is ten times the place before it. So ten ones make a ten, and ten tens make a hundred. Historians are not sure exactly where he was born. He wrote his numbers using letters, not the digits we use today."

DO Then ask: "Would anyone like to try this one?" Try it: Aryabhata's rule: each place is ten times the place before it. In the number 538, which digit is in the tens place, and what is it worth? What is ten times 50? Volunteers only, never name a child, nobody is marked, passing is fine. SKIP the story break, without comment, if you are behind time or the group needs the full four minutes.

36–72 min Worksheet (pairs, then alone, then stretch)

DO Minute 36: give out the workbook. Read the title of Lesson 1 aloud. Say "Pairs first. Talk before you write."

Minutes 36 to 50: pairs work together on the first page (online, visit every shared breakout room by minute 47 so no pair works unseen for long; in Lessons 5, 6 and 10 a short teach slide comes first, 36 to 38). Minutes 50 to 52: check page 1 together. Minutes 52 to 57: short teaching for page 2. Minutes 57 to 69: each child works alone on the second page. Minutes 69 to 72: check page 2 together. Think harder, Go further and the Investigation are open to any child at any time after question 3.

SAY *"Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking."*

DO Circulate and ask:

- Ask: "How many of your numbers start with this card? Is that all of them?"
- Ask: "What is this digit worth? How many tens is that?"
- Ask: "Point to the number one square below this one on the chart. How much bigger is it?"

IF STUCK Hint 1: "Fix the hundreds digit first."

Hint 2: "Build it with blocks: hundreds, tens and ones. Can 10 tens become a hundred?"

Hint 3: "Say the numbers out loud as you count on in 10s. What happens after 590?"

DO A choice, not a track: after question 3 any child may carry on or try Think harder (the neutral line is printed on page 1). Invite the WHOLE group at once, never one child: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Never announce who is on which page, never name a child in front of the group, and do not split a pair unless both children agree. Children may move either way at any time. The Investigation is printed on the last core page, so every child sees it.

Think harder and Go further questions

Question	Answer
Workbook Think harder 1: Use the digit cards 1, 5 and 8 once each. How many different 3-digit numbers can you make? List them in order.	6 numbers: 158, 185, 518, 581, 815, 851.
Workbook Think harder 2: Use the digit cards 4, 4 and 7. How many different 3-digit numbers can you make? Why is it not 6?	3 numbers: 447, 474, 744. The two 4s look the same, so swapping them does not make a new number.
Workbook Think harder 3: What number is 3 hundreds, 14 tens and 6 ones?	446 (14 tens = 1 hundred and 4 tens).
Workbook Think harder 4: Mei says: "If I swap the tens digit and the ones digit of a 3-digit number, the new number is always smaller." Is she right? Show what happens with some examples. When does the number stay the same?	No. 452 becomes 425, which is smaller, but 425 becomes 452, which is bigger. It gets smaller when the tens digit is bigger than the ones digit, bigger when it is smaller, and stays the same when the two digits are the same (733).
Workbook Go further 5: Mei uses each of the digit cards 1, 2, 3, 4, 5 and 6 once to make two 3-digit numbers. She adds them. What is the smallest total she can make?	381, for example 135 + 246. The smallest digits 1 and 2 go in the hundreds places, 3 and 4 in the tens, 5 and 6 in the ones: 300 + 70 + 11 = 381.
Workbook Go further 6: Ali counts in 5s, starting at 3: 3, 8, 13, 18 … Will he say 500? Will he say 998? Explain.	He says only numbers ending in 3 or 8. So he will not say 500, but he will say 998.

Workbook Go further 7: Ben writes all the numbers from 300 to 399. How many times does he write the digit 3?	120. Hundreds place: 100 times. Tens place (330 to 339): 10 times. Ones place (303, 313, ... 393): 10 times. 100 + 10 + 10 = 120.

These are the questions on the two Think harder pages in the workbook (Think harder, then Go further). More are in the separate Question Bank.

Investigation (several answers are right)	Model answer and what to look for
Workbook Investigation (last core page): Mirror numbers. A mirror number reads the same forwards and backwards, like 252 or 909. (a) Write 5 mirror numbers of your own. (b) Find ALL the mirror numbers from 200 to 299. How do you know you have found them all? (c) Ali says every mirror number is odd. Is he right? Easy first step: write 252, 909 and 131, then write three more mirror numbers.	(a) Any number whose first and last digits match, such as 101, 454, 777. (b) 10 numbers: 202, 212, 222, 232, 242, 252, 262, 272, 282, 292. The first and last digits must both be 2, and the middle digit can be 0 to 9: 10 choices. A child who lists them by the middle digit in order has not missed any. (c) No: 202 is even. A mirror number is even when its first (and last) digit is even. Extension: there are 90 mirror numbers from 100 to 999 (9 choices for the end digits x 10 for the middle). Look for: the child sees that the end digits match, lists in order, and can say why the list is complete.

72–77 min Talk time

DO Bring the group back to a circle (or camera on, mics on). You choose volunteers first: "Who would like to share?" If nobody offers, ask the pairs to share what their partner said. Never cold-call a child by name. Any child may say "pass" at any time and that is always fine. Do not keep public tallies, scores or rankings (no counting who got most right, no "first to finish"). Invite the whole group to think about whether they would like a turn, never one child.

SAY *"Which is bigger, 401 or 399? Ali says 399 because it has two 9s. Who agrees?"*

"How do you know you found every number from your digit cards?"

Discussion question	Listen for
Which is bigger, 401 or 399? Ali says 399 because it has two 9s. Who is right, and why?	401 is bigger. Compare the hundreds first: 4 hundreds is more than 3 hundreds, so the tens and ones do not matter. We know we have every number because we listed them in order: each card starts the same number of numbers.

77–80 min Investigation share (3 minutes)

DO Show the Investigation slide. The Investigation is on the last core page of the workbook and is open to every child from question 3 on. Ask the whole group: "Who would like to share one way they tried it? Passing is fine." Volunteers only, no names, no public tallies. Several answers are right: praise the method, not the speed. If nobody has started, give pairs one minute for the easy first step and set the rest as take-home.

Investigation	Model answer and what to look for

Investigation: Mirror numbers	(a) Any number whose first and last digits match, such as 101, 454, 777. (b) 10 numbers: 202, 212, 222, 232, 242, 252, 262, 272, 282, 292. The first and last digits must both be 2, and the middle digit can be 0 to 9: 10 choices. A child who lists them by the middle digit in order has not missed any. (c) No: 202 is even. A mirror number is even when its first (and last) digit is even. Extension: there are 90 mirror numbers from 100 to 999 (9 choices for the end digits x 10 for the middle). Look for: the child sees that the end digits match, lists in order, and can say why the list is complete.

80–87 min Exit ticket

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide.

Exit ticket	Answer
Use the digit cards 6, 0 and 2 once each. Write the biggest and the smallest 3-digit number.	Biggest 620, smallest 206.

87–89 min Looking back (2 minutes)

SAY *"Looking back takes 2 minutes. Find the small box called Looking back on your page. Finish one line, two lines or all three, and tick what helped you. You may pass on any line. This box is for you. If you tick "I am still stuck", your tutor will read it at the next lesson. Nothing is shared with the group."*

DO Nothing is collected in front of the group and nobody has to read theirs out. The Looking back box is for the child. A paper box cannot be read live, so the tutor reads the journals at the next lesson, while stamping, and never by private message or photo. If a child ticked still stuck, start the next open session with a whole-group invitation (for example: "Anyone who would like another look at last lesson's idea, stay with me and we will do it together"), with a parent nearby; never single out one child. A child may volunteer one sentence, never by name unless they choose to. No public tallies. Praise the strategy, not the speed.

89–90 min Closing and stamps

SAY *"You were real detectives today: you made a list in order, so no number could hide from you."*

"Hand me your exit ticket and I will stamp your inquiry journal."

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide (this is for you only, never shared with the children). If a child marked D or H, plan two minutes of catch-up at the start of the next lesson.

Think harder and Go further

Use any three of the cards 1, 3, 5 and 7. How many different 3-digit numbers can you make? (24: 6 for each set of three cards, and there are 4 sets.)

Answer key (workbook)

Q1 (a) 407 (b) 752 (c) 3 tens; 13 tens (d) 70

Q2 (a) Start with 2: 247, 274. Start with 4: 427, 472. Start with 7: 724, 742. (b) Biggest 742, smallest 247 (c) 472 (only 28 away from 500) (d) 2 odd numbers: 247 and 427

Q3 (a) 509, 590, 905, 950 (b) 0 cannot be the hundreds digit (059 is not a 3-digit number), so only 5 or 9 can start: 2 + 2 = 4 numbers, not 6.

Q4 Piece A: 348, 349 / 357, 359 / 367, 368. Piece B: 595, 596, 597 in the middle row and 606 at the bottom. (a) Down: 10 more. Right: 1 more. (b) 405 (395, then 405)

Q5 (a) 347, 447 (b) 502, 492 (c) 1000 (d) 700, 699

Q6 (a) 435 (b) 6 numbers: 102, 111, 120, 201, 210, 300

Story try-it (Aryabhata). The digit 3 is in the tens place and is worth 30. Ten times 50 is 500 (which is why the 5 in 538 is worth 500).

Investigation (Mirror numbers). (a) Any number whose first and last digits match, such as 101, 454, 777. (b) 10 numbers: 202, 212, 222, 232, 242, 252, 262, 272, 282, 292. The first and last digits must both be 2, and the middle digit can be 0 to 9: 10 choices. A child who lists them by the middle digit in order has not missed any. (c) No: 202 is even. A mirror number is even when its first (and last) digit is even. Extension: there are 90 mirror numbers from 100 to 999 (9 choices for the end digits x 10 for the middle). Look for: the child sees that the end digits match, lists in order, and can say why the list is complete.

Lesson 2: Missing Digit Mysteries

Week 1 (second lesson) • Addition and Subtraction within 1000 • Workbook: Lesson 2

Objective	Find missing digits in 3-digit addition and subtraction by starting at the ones and checking for a carry, use "=" as "is the same as" to balance number sentences, and complete magic squares and triangles with bigger numbers.
Thinking skills	Working backwards, guess and check, reasoning about regrouping, balancing, drawing a model
Materials	Digit cards 0 to 9, base-ten blocks or place-value counters, mini whiteboards, a paper magic triangle with 6 circles and number tiles 110 to 160.
Online set-up	Children write column sums on whiteboards or paper and show your tutor on camera or keep it to share if you want; number tiles are cut from paper before the lesson. The tutor writes on the slide sums.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0–1 min Opening

SAY *"Welcome back, detectives! Someone has spilled ink on our sums and some digits have disappeared."*

"Today we find the missing digits, make both sides of "=" balance, and crack some magic squares. Notice it. Test it. Explain why."

DO Greet each child by name. Show the lesson title slide. Read "Today we will..." aloud.

1–4 min Sneak-peek puzzles (3 minutes)

SAY *"These are sneak-peek puzzles. They are ideas we have not learnt yet, so guessing is welcome and so is saying "I do not know". Nobody is marked and nothing is read out. Start with question 1: use objects, a drawing or counting. Then try question 2. Show your answer on your own paper to your tutor on camera, or keep it to share if you want. If you would like help, the tutor will invite the whole group to ask in the open session, and a parent can stay nearby."*

DO Show the Sneak-peek puzzles slide at minute 1 (3 minutes: about 1 minute for question 1, then 2 minutes for question 2). Say out loud: these are ideas we have not learnt yet; guessing and "I do not know" are welcome; nobody is marked.

Answers stay private. Each child holds their own paper to the camera for you only (look, do not read out), or writes it on their own paper and shows only you on camera. Never raise whiteboards together and never show a board in front of the group. Never ask a named child "how do you know?" Ask for volunteers: "Would anyone like to tell us how they started?"

Question 1 is an access question: every child can try it with objects, a drawing or counting, so everyone has a first win. Question 2 is the thinking question. Do not mark either one. Click to show

the answers at the end as "one way to think about it", without saying who had what.

Use what you see only to plan. Invite the WHOLE group at once, never one child: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Never announce who is on which page and never name a child in front of the group. Children choose, and may move either way at any time. Pairs are not split unless both children agree. Everybody joins the explore, the mini-lesson, the check-together slides and talk time.

Sneak-peek puzzle	Answer and what to listen for
Question 1 (access: objects, drawing or counting): 3[ ]2 + 416 = 788. Which digit is hidden in the box? (Use counters or draw the columns if it helps.)	7 (372 + 416 = 788). Listen for: No carry in this one: look at the tens column, 1 and what makes 8. This is a first win for everyone.
Question 2 (thinking): 4[ ]6 + 237 = 683. Which digit is hidden in the box? How do you know?	4 (446 + 237 = 683). Listen for: Ones: 6 + 7 = 13, so 1 ten is carried. Tens: the box + 3 + 1 must end in 8, so the box is 4. Listen for: "carry", or a guess that was tested and checked.

DO The neutral line “After question 3 you may choose: carry on, or try Think harder.” is printed on page 1 of the workbook, next to the lesson title. Nobody is labelled and nothing is announced.

4–8 min Warm-up

DO Partners to 100 and 1000. The tutor says a number; say its partner. 65 → 35, then 370 → 630, then 125 → 875.

SAY *“I say a number. You say its partner to make 100. Ready? 65!”*

“Now partners to 1000. 370! How did you find 630 so fast?”

DO Keep the game to two minutes (4 to 6). Then the quick questions (6 to 8): ask questions 1, 3 and 5; questions 2 and 4 are spare. Children answer by thumbs, a whisper to a partner. Do not name a child.

Ask	Expect
65 + __ = 100	35
300 + 450 = ?	750
700 – 250 = ?	450
238 + 10 = ?	248
500 – __ = 120	380

8–20 min Explore (in pairs, with objects)

SAY *“Pick any 3-digit number. I predict that its partner to make 999 will be easy to find, with no carrying at all. Do you think I am right? Let us test it.”*

DO 1. Pick any 3-digit number, like 358.

2. Find its partner that makes 999. Check by adding.

3. Do it for three more numbers.

4. Look at each column. What do you notice?

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK Stuck: start with the hundreds. What goes with 3 to make 9?

Not sure: add your two numbers in a column sum to check the total is 999.

Finished early: try partners to 1000. What changes in the ones column?

DO Go further (a second what-if, so nobody waits): show the Explore: go further slide or post it in the chat when pairs go to their rooms. Any child may try it any time, as a choice. Never name who does. Question: now make 1000. What is the partner of 358 that makes 1000 (358 + ? = 1000)? What do you notice about the columns? Answer: 642. Ones: 8 + 2 = 10. Tens: 5 + 4 + 1 = 10. Hundreds: 3 + 6 + 1 = 10. Every column carries.

DO At about minute 14, say "Pencils down" and bring the group back together. Take the three questions below, one at a time (about 6 minutes).

Ask	Expect
What did you notice about each column?	The two digits in each column add up to 9.
Why is there never a carry?	No column adds up to more than 9.
Find the partner of 274 without adding.	725: 2 + 7, 7 + 2 and 4 + 5 all make 9.

20–32 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"4A7 + 28B = 723. I start at the ones: 7 plus B ends in 3. 7 + 6 = 13. So B is 6, and I carry 1 ten."*

"Tens: A + 8 + the carried 1 ends in 2. 3 + 8 + 1 = 12. So A is 3, and I carry 1 hundred. Hundreds: 4 + 2 + 1 = 7. It matches! I check by adding: 437 + 286 = 723."

"6C2 − 24D = 385. Subtraction is harder, so I work backwards and turn it into an addition: 385 + 24D = 6C2. Ones: 5 + D ends in 2, 5 + 7 = 12, so D is 7. Tens: 8 + 4 + 1 = 13, so C is 3. Hundreds: 3 + 2 + 1 = 6 ✓."

"245 + 130 = 200 + E. I do not need to add. 245 is 45 more than 200, so E must be 45 more than 130. That is 175. Both sides are 375."

Worked example on the slides	Answer

Missing digits: start at the ones. 4A7 + 28B = 723 Ones: 7 + B ends in 3, so B = 6 (13, carry 1). Tens: A + 8 + 1 ends in 2, so A = 3 (12, carry 1). Hundreds: 4 + 2 + 1 = 7 ✓	437 + 286 = 723.
Balance: "=" means "is the same as". 245 + 130 = 200 + E 245 is 45 more than 200, so E is 45 more than 130.	175. Both sides are 375.

SAY *"The key idea today is: start at the ones. Start at the ones column and ask: was there a carry? Check by adding. Say it with me."*

DO Show the key idea slide at about minutes 31 to 32. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Forgetting the carried 1: in Q1c writing 6 in the tens box (8 + 6 = 14) instead of 5 (8 + 5 + 1 = 14).
- Reading "=" as "the answer comes next": 350 + 140 = 300 + A answered 490.
- Magic triangle: putting the biggest numbers in the corners when the side total is small (390 needs the smallest corners).

32–36 min Brain break

DO Balance bodies. Stand like a see-saw with arms out. The tutor says a number sentence: stay level if it is true, tip to one side if it is false.

Water. Toilet break for anyone who needs it. Keep it to four minutes.

DO Story break (optional, 2 minutes): do the movement game for 2 minutes (32 to 34), then show the "Story break: Carl Friedrich Gauss" slide (34 to 36). Read the story aloud (about 30 seconds): "A story often told about young Gauss goes like this. At school in Brunswick, his teacher asked the class to add up the numbers from 1 to 100. Gauss found the answer, 5050, very quickly. He paired 1 with 100, 2 with 99, and so on. Every pair makes 101, and there are 50 pairs. Historians are not sure it happened exactly like this. The oldest written version came after he died and does not give the numbers. But pairing numbers to add faster is real, clever maths!" Say clearly: A story often told. Historians are not sure it happened exactly like this.

DO Then ask: "Would anyone like to try this one?" Try it: Use Gauss's pairing trick to add 1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10. Which pairs do you make? Volunteers only, never name a child, nobody is marked, passing is fine. SKIP the story break, without comment, if you are behind time or the group needs the full four minutes.

36–72 min Worksheet (pairs, then alone, then stretch)

DO Minute 36: give out the workbook. Read the title of Lesson 2 aloud. Say "Pairs first. Talk before you write."

Minutes 36 to 50: pairs work together on the first page (online, visit every shared breakout room by minute 47 so no pair works unseen for long; in Lessons 5, 6 and 10 a short teach slide comes first, 36 to 38). Minutes 50 to 52: check page 1 together. Minutes 52 to 57: short teaching for page 2. Minutes 57 to 69: each child works alone on the second page. Minutes 69 to 72: check page 2

together. Think harder, Go further and the Investigation are open to any child at any time after question 3.

SAY *"Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking."*

DO Circulate and ask:

- Ask: "Which column did you start with? Why?"
- Ask: "Was there a carry into this column? How do you know?"
- Ask: "What is each side of the = sign worth? Are they the same?"

IF STUCK Hint 1: "Start at the ones column."

Hint 2: "Try a digit in the box and add. Too big or too small?"

Hint 3: "Turn the subtraction into an addition: answer + bottom number = top number."

DO A choice, not a track: after question 3 any child may carry on or try Think harder (the neutral line is printed on page 1). Invite the WHOLE group at once, never one child: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Never announce who is on which page, never name a child in front of the group, and do not split a pair unless both children agree. Children may move either way at any time. The Investigation is printed on the last core page, so every child sees it.

Think harder and Go further questions

Question	Answer
Workbook Think harder 1: A magic triangle has 3 corner circles and 3 circles in the middle of the sides. Put the numbers 110, 120, 130, 140, 150 and 160 in the circles, once each, so that every side adds up to 420. Draw the triangle. Which numbers are in the corners?	Corners 140, 150, 160. Between 140 and 150: 130. Between 150 and 160: 110. Between 160 and 140: 120 (each side: 140 + 130 + 150 = 420, 150 + 110 + 160 = 420, 160 + 120 + 140 = 420).
Workbook Think harder 2: Find the missing digits: 5A7 + B48 = 905	557 + 348 = 905 (ones 7 + 8 = 15; tens 5 + 4 + 1 = 10; hundreds 5 + 3 + 1 = 9).
Workbook Think harder 3: Find the missing digits: 7A1 − B58 = 263	721 − 458 = 263. Check: 263 + 458 = 721.
Workbook Think harder 4: What number makes this true? 538 + 199 = 540 + A	197. 538 is 2 less than 540, so A is 2 less than 199. Both sides are 737.
Workbook Go further 5: Find ALL the ways to fill in the missing digits: AB7 + 4CD = 1000	10 ways: 507 + 493, 517 + 483, 527 + 473, 537 + 463, 547 + 453, 557 + 443, 567 + 433, 577 + 423, 587 + 413, 597 + 403. Ones: 7 + 3 = 10. Hundreds: 5 + 4 + 1 = 10. The two tens digits add up to 9 (with the carried 1 they make 10).

Workbook Go further 6: Box A and Box B hold 250 marbles together. Box B and Box C hold 350. Box A and Box C hold 400. How many marbles are in each box?	A 150, B 100, C 250. B + C is 100 more than A + B, so C has 100 more than A. A + C = 400, so A = 150 and C = 250. Then B = 250 − 150 = 100.
Workbook Go further 7: Use the digit cards 1, 2, 3, 4, 5 and 6 once each to make two 3-digit numbers. Make the difference as SMALL as possible.	47: 412 − 365. The hundreds digits are next to each other (4 and 3); then make the number starting with 4 as small as possible (412) and the one starting with 3 as big as possible (365).

These are the questions on the two Think harder pages in the workbook (Think harder, then Go further). More are in the separate Question Bank.

Investigation (several answers are right)	Model answer and what to look for
Workbook Investigation (last core page): Odd or even?. Ali says: "When I add two odd numbers, I always get an even number." (a) Test Ali's idea with 5 pairs of odd 3-digit numbers (for example 345 + 271). (b) Now test odd + even, and even + even. What do you notice? (c) Explain WHY, using pairs of counters or a drawing. Easy first step: use counters to try 3 + 5, 7 + 9 and 11 + 13. Are the answers odd or even?	(a) Ali is right: odd + odd is always even (checked for every pair of 3-digit numbers). (b) Odd + even is always odd; even + even is always even. (c) Why: an odd number is some pairs plus 1 left over, an even number is only pairs. Two odd numbers have two left-overs, and they make one more pair, so no one is left over. Look for: several tests before a claim, then a reason that uses pairs, counters or ones digits, not only "it worked every time". A ones-digit reason (odd ends in 1, 3, 5, 7, 9) is also fine. Extension (not printed): taking away behaves the same way: odd − odd is even, odd − even is odd.

72–77 min Talk time

DO Bring the group back to a circle (or camera on, mics on). You choose volunteers first: "Who would like to share?" If nobody offers, ask the pairs to share what their partner said. Never cold-call a child by name. Any child may say "pass" at any time and that is always fine. Do not keep public tallies, scores or rankings (no counting who got most right, no "first to finish"). Invite the whole group to think about whether they would like a turn, never one child.

SAY *"Why do we start at the ones column when we look for missing digits?"*

"Ravi says 398 + 205 = 400 + 203 without adding. How can he know?"

Discussion question	Listen for
Why do we start at the ones column when we look for missing digits?	The ones column can make a carry that changes the tens column, and the tens can change the hundreds, so we must know the ones first. Ravi moved 2 from 205 onto 398: the total does not change, so both sides are 603.

77–80 min Investigation share (3 minutes)

DO Show the Investigation slide. The Investigation is on the last core page of the workbook and is open to every child from question 3 on. Ask the whole group: “Who would like to share one way they tried it? Passing is fine.” Volunteers only, no names, no public tallies. Several answers are right: praise the method, not the speed. If nobody has started, give pairs one minute for the easy first step and set the rest as take-home.

Investigation	Model answer and what to look for
Investigation: Odd or even?	(a) Ali is right: odd + odd is always even (checked for every pair of 3-digit numbers). (b) Odd + even is always odd; even + even is always even. (c) Why: an odd number is some pairs plus 1 left over, an even number is only pairs. Two odd numbers have two left-overs, and they make one more pair, so no one is left over. Look for: several tests before a claim, then a reason that uses pairs, counters or ones digits, not only "it worked every time". A ones-digit reason (odd ends in 1, 3, 5, 7, 9) is also fine. Extension (not printed): taking away behaves the same way: odd − odd is even, odd − even is odd.

80–87 min Exit ticket

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide.

Exit ticket	Answer
3A4 + 257 = 601. Which digit is A?	A = 4 (344 + 257 = 601: ones 4 + 7 = 11; tens 4 + 5 + 1 = 10). Bonus for anyone who would like it: 4A6 + 37B = 801 gives A = 2 and B = 5 (426 + 375 = 801).

87–89 min Looking back (2 minutes)

SAY *“Looking back takes 2 minutes. Find the small box called Looking back on your page. Finish one line, two lines or all three, and tick what helped you. You may pass on any line. This box is for you. If you tick "I am still stuck", your tutor will read it at the next lesson. Nothing is shared with the group.”*

DO Nothing is collected in front of the group and nobody has to read theirs out. The Looking back box is for the child. A paper box cannot be read live, so the tutor reads the journals at the next lesson, while stamping, and never by private message or photo. If a child ticked still stuck, start the next open session with a whole-group invitation (for example: "Anyone who would like another look at last lesson's idea, stay with me and we will do it together"), with a parent nearby; never single out one child. A child may volunteer one sentence, never by name unless they choose to. No public tallies. Praise the strategy, not the speed.

89–90 min Closing and stamps

SAY *"You were digit detectives again: you started at the ones, checked every carry and proved each answer by adding."*

"Exit ticket, then your stamp."

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide (this is for you only, never shared with the children). If a child marked D or H, plan two minutes of catch-up at the start of the next lesson.

Think harder and Go further

Make your own missing-digit sum with two boxes that has exactly ONE answer. Swap with a partner. (Check: does any other digit also work?)

Answer key (workbook)

Q1 (a) 224 + 355 = 579 (b) 462 + 375 = 837 (c) 386 + 259 = 645

Q2 (a) 778 – 325 = 453 (b) 543 – 267 = 276

Q3 (a) 190 (b) 525 (c) 280 (d) Yes. 398 is 2 less than 400 and 205 is 2 more than 203, so both sides are the same (603).

Q4 (a) 150 (diagonal 80 + 50 + 20) (b) Top: 10. Middle: 30 and 70. Bottom: 90. (c) Yes, still magic. New total 450: every line has 3 numbers and each is 100 more, so 150 + 300.

Q5 (a) Corners 110, 120, 130. Between 110 and 120: 160. Between 120 and 130: 140. Between 130 and 110: 150. (Think harder 1 is the same triangle with sides of 420: corners 140, 150, 160; between 140 and 150: 130; between 150 and 160: 110; between 160 and 140: 120.)

Q6 250 and 450 (700 – 200 = 500, half of 500 is 250, 250 + 200 = 450).

Story try-it (Carl Friedrich Gauss). Pairs 1+10, 2+9, 3+8, 4+7, 5+6 each make 11. There are 5 pairs, so 5 × 11 = 55.

Investigation (Odd or even?). (a) Ali is right: odd + odd is always even (checked for every pair of 3-digit numbers). (b) Odd + even is always odd; even + even is always even. (c) Why: an odd number is some pairs plus 1 left over, an even number is only pairs. Two odd numbers have two left-overs, and they make one more pair, so no one is left over. Look for: several tests before a claim, then a reason that uses pairs, counters or ones digits, not only "it worked every time". A ones-digit reason (odd ends in 1, 3, 5, 7, 9) is also fine. Extension (not printed): taking away behaves the same way: odd – odd is even, odd – even is odd.

Lesson 3: Legs, Wheels and Arrays

Week 2 (first lesson) • Multiplication and Division • Workbook: Lesson 3

Objective	Read an array as rows of equal groups and write two multiplication and two division sentences for it; solve "how many of each" legs and wheels puzzles with a table; share with leftovers and decide what the leftover means; find all the arrays for a number.
Thinking skills	Draw it (arrays), make a table, guess and check, working in order to find all the ways
Materials	About 30 counters per pair, mini whiteboards, squared paper. Optional: picture cards of hens and goats, bicycles and tricycles.
Online set-up	Each child has 30 small objects (buttons, dried beans or Lego bricks) and a whiteboard or paper. The tutor shares the slides; children show their tutor on camera or keep it to share if you want (arrays and tables).
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0–1 min Opening

SAY *"Welcome back, inquirers! Today we are on a farm and in a bicycle shop. We will count legs and wheels, line things up in neat rows called arrays, and share things out fairly."*

"Our big question today: when there are two kinds of things mixed up, how can we find out how many of each there are, without guessing wildly?"

DO Greet each child by name. Show the lesson title slide. Read "Today we will..." aloud.

1–4 min Sneak-peek puzzles (3 minutes)

SAY *"These are sneak-peek puzzles. They are ideas we have not learnt yet, so guessing is welcome and so is saying "I do not know". Nobody is marked and nothing is read out. Start with question 1: use objects, a drawing or counting. Then try question 2. Show your answer on your own paper to your tutor on camera, or keep it to share if you want. If you would like help, the tutor will invite the whole group to ask in the open session, and a parent can stay nearby."*

DO Show the Sneak-peek puzzles slide at minute 1 (3 minutes: about 1 minute for question 1, then 2 minutes for question 2). Say out loud: these are ideas we have not learnt yet; guessing and "I do not know" are welcome; nobody is marked.

Answers stay private. Each child holds their own paper to the camera for you only (look, do not read out), or writes it on their own paper and shows only you on camera. Never raise whiteboards together and never show a board in front of the group. Never ask a named child "how do you know?" Ask for volunteers: "Would anyone like to tell us how they started?"

Question 1 is an access question: every child can try it with objects, a drawing or counting, so everyone has a first win. Question 2 is the thinking question. Do not mark either one. Click to show the answers at the end as "one way to think about it", without saying who had what.

Use what you see only to plan. Invite the WHOLE group at once, never one child: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Never announce who is on which page and never name a child in front of the group. Children choose, and may move either way at any time. Pairs are not split unless both children agree. Everybody joins the explore, the mini-lesson, the check-together slides and talk time.

Sneak-peek puzzle	Answer and what to listen for
Question 1 (access: objects, drawing or counting): A tray has 3 rows of counters with 5 counters in each row. Draw it if you like. How many counters are there?	15 counters (3 rows of 5: 5, 10, 15). Listen for: Counting in 5s, or drawing and counting. This is a first win for everyone.
Question 2 (thinking): A toy shop has cars (4 wheels) and scooters (2 wheels). 6 toys have 16 wheels altogether. How many cars and how many scooters? How do you know?	2 cars and 4 scooters (8 + 8 = 16). Listen for: Starting with all scooters (12 wheels) and swapping one at a time is the idea of today. A child who draws 6 vehicles and counts wheels is thinking well. Guess and check is welcome.

DO The neutral line “After question 3 you may choose: carry on, or try Think harder.” is printed on page 1 of the workbook, next to the lesson title. Nobody is labelled and nothing is announced.

4–8 min Warm-up

DO Legs and wheels count. Picture cards: 4 cars, 5 bicycles, 3 tricycles, 5 dogs, 6 hens, 2 cars and 1 bicycle. Count the wheels or legs in 2s, 3s, 4s or 5s.

SAY *“Here are some cards. Each card shows some cars, bicycles or animals. Count the wheels or the legs. Count in 2s, 3s, 4s or 5s, not in 1s!”*

“Show me the number with your fingers or on your whiteboard. Then we will click to check.”

DO Keep the game to two minutes (4 to 6). Then the quick questions (6 to 8): ask questions 1, 3 and 5; questions 2 and 4 are spare. Children answer by thumbs, a whisper to a partner. Do not name a child.

Ask	Expect
3 × 5 = ?	15
4 groups of 4. How many?	16
20 ÷ 2 = ?	10
30 shared equally by 10 children. How many each?	3
How many 5s make 35?	7

8–20 min Explore (in pairs, with objects)

SAY *“The farmer has only hens and goats in a pen. He counts 16 legs. How many animals do you think there are? Is there only one answer? Write a guess.”*

DO 1. A pen has only hens (2 legs) and goats (4 legs).

2. The farmer counts 16 legs altogether.

3. Use counters or drawings to find a mix of hens and goats.

4. Find every mix. Record each one.

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK Stuck: try all hens first. How many hens make 16 legs?

Then swap one hen for a goat. What happens to the number of legs? Fix it.

Finished early: which mix has the most animals? Which has the fewest? How do you know there are no more mixes?

DO Go further (a second what-if, so nobody waits): show the Explore: go further slide or post it in the chat when pairs go to their rooms. Any child may try it any time, as a choice. Never name who does. Question: what if the farmer counts 20 legs? How many different mixes of hens and goats are there now? Answer: 6 mixes: 10 hens; 8 hens and 1 goat; 6 hens and 2 goats; 4 hens and 3 goats; 2 hens and 4 goats; 5 goats.

DO At about minute 14, say "Pencils down" and bring the group back together. Take the three questions below, one at a time (about 6 minutes).

Ask	Expect
How many different mixes did you find?	5 mixes: 8 hens; 6 hens and 1 goat; 4 hens and 2 goats; 2 hens and 3 goats; 4 goats.
How do you know you have them all?	The goats went 0, 1, 2, 3, 4. Five goats would need 20 legs, too many.
What happens each time you swap 2 hens for 1 goat?	The legs stay at 16: 2 hens have 4 legs, the same as 1 goat.

20–32 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"3 rows of 4 dots. I count the rows: 3. I count how many in each row: 4. 4, 8, 12. So 3 × 4 = 12. If I turn my head, it is 4 rows of 3, so 4 × 3 = 12 too. Sharing 12 into 3 rows gives 4 in each row: 12 ÷ 3 = 4. Putting 12 in groups of 4 gives 3 groups: 12 ÷ 4 = 3."*

"Part of this tray is under a cloth. I do not need to see every cookie. The top row shows 5 in each row. The side shows 4 rows. 4 rows of 5 is 20. I can see 8, so 20 – 8 = 12 are hidden."

"6 vehicles, bicycles and cars, 16 wheels. I start with all bicycles: 6 bicycles is 12 wheels. Too few. I swap one bicycle for a car: that adds 2 wheels, 14. Swap again: 16. So 4 bicycles and 2 cars. Check: 8 + 8 = 16 wheels, 4 + 2 = 6 vehicles."

Worked example on the slides	Answer

Arrays: rows of equal groups. 3 rows of 4 dots. 3 × 4 = 12 and 4 × 3 = 12. 12 ÷ 3 = 4 and 12 ÷ 4 = 3. A tray of 4 rows of 5 cookies has some under a cloth. We can see 8. How many are hidden?	12 hidden (4 rows of 5 is 20, and 20 − 8 = 12).
How many of each? Make a table. 6 vehicles: bicycles (2 wheels) and cars (4 wheels). 16 wheels altogether. Start with all bicycles: 6 bicycles, 12 wheels. Swap one bicycle for a car each time: 14, then 16 wheels. How many of each?	4 bicycles and 2 cars (8 + 8 = 16 wheels).

SAY *"The key idea today is: draw it, then make a table. An array is rows of equal groups: rows × number in each row. For "how many of each", start with all of one kind and swap one at a time in a table. Say it with me."*

DO Show the key idea slide at about minutes 31 to 32. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Counting the hidden part of an array one by one (or guessing) instead of using rows × number in each row and taking away the ones you can see.
- Giving the number of whole groups as the answer when the leftover still needs a place (28 children, vans of 5: writing 5 vans instead of 6).
- In "how many of each" puzzles, matching only one clue (the legs) and forgetting to check the other (the number of animals).

32–36 min Brain break

DO Array moves. The tutor calls "3 rows of 4!": do 3 rows of 4 claps, counting 4, 8, 12. Then "5 rows of 2" jumps and "2 rows of 10" stomps.

Water. Toilet break for anyone who needs it. Keep it to four minutes.

DO Story break (optional, 2 minutes): do the movement game for 2 minutes (32 to 34), then show the "Story break: Shakuntala Devi" slide (34 to 36). Read the story aloud (about 30 seconds): "Shakuntala Devi was born in Bangalore, India, in 1929. Her father saw how good she was at numbers when she was very young. At six she showed her sums at a university. In 1980, in London, two 13-digit numbers were picked for her at random. She multiplied them in her head and said the answer in 28 seconds! That feat was written in the Guinness Book of World Records. People called her the human computer."

DO Then ask: "Would anyone like to try this one?" Try it: Do these in your head like Shakuntala: 7 × 5, 9 × 4 and 6 × 10. Which answer is the biggest, and which is the smallest? Volunteers only, never name a child, nobody is marked, passing is fine. SKIP the story break, without comment, if you are behind time or the group needs the full four minutes.

36–72 min Worksheet (pairs, then alone, then stretch)

DO Minute 36: give out the workbook. Read the title of Lesson 3 aloud. Say "Pairs first. Talk before you write."

Minutes 36 to 50: pairs work together on the first page (online, visit every shared breakout room by minute 47 so no pair works unseen for long; in Lessons 5, 6 and 10 a short teach slide comes first, 36 to 38). Minutes 50 to 52: check page 1 together. Minutes 52 to 57: short teaching for page 2. Minutes 57 to 69: each child works alone on the second page. Minutes 69 to 72: check page 2 together. Think harder, Go further and the Investigation are open to any child at any time after question 3.

SAY *"Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking."*

DO Circulate and ask:

- Ask: "How many rows? How many in each row? Which multiplication is that?"
- Ask: "What changes each time you swap one animal for the other kind?"
- Ask: "Did you check BOTH clues: the number of animals and the number of legs?"

IF STUCK Hint 1: "Draw it. Dots in rows, or circles for heads and sticks for legs."

Hint 2: "Start with all of one kind. Then swap one at a time."

Hint 3: "Make a table so you can see the pattern and do not miss a row."

DO A choice, not a track: after question 3 any child may carry on or try Think harder (the neutral line is printed on page 1). Invite the WHOLE group at once, never one child: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Never announce who is on which page, never name a child in front of the group, and do not split a pair unless both children agree. Children may move either way at any time. The Investigation is printed on the last core page, so every child sees it.

Think harder and Go further questions

Question	Answer
Workbook Think harder 1: A shop has some bicycles (2 wheels) and some tricycles (3 wheels), at least one of each. There are 17 wheels altogether. Find ALL the ways. Start with 1 tricycle, then 2 tricycles, and so on. Make a table or use counters. How do you know you have found them all?	3 ways: 7 bicycles and 1 tricycle (14 + 3); 4 bicycles and 3 tricycles (8 + 9); 1 bicycle and 5 tricycles (2 + 15). 0, 2 or 4 tricycles leave an odd number of wheels, which bicycles cannot make, and 6 or more tricycles need 18 wheels or more.
Workbook Think harder 2: There are 9 hens and goats on a farm. They have 24 legs altogether. How many hens and how many goats are there?	6 hens and 3 goats. Start with 9 hens (18 legs); each swap to a goat adds 2 legs; 3 swaps make 24. Check: 12 + 12 = 24.
Workbook Think harder 3: Ali shares 34 marbles equally among 4 friends, giving as many as he can. How many does each friend get? How many are left over?	8 each (4 × 8 = 32) and 2 left over.

Workbook Think harder 4: An array of stickers has 4 rows. Ben counts 12 stickers in the first 3 rows. How many stickers are in the whole array?	16 stickers. 3 rows have 12, so each row has 4 (12 ÷ 3 = 4); 4 rows of 4 is 16.
Workbook Go further 5: A toy shop has tricycles (3 wheels) and toy cars (4 wheels), at least one of each. There are 30 wheels altogether. Find all the ways.	2 ways: 6 tricycles and 3 cars (18 + 12); 2 tricycles and 6 cars (6 + 24). With 1, 2, 4, 5 or 7 cars the wheels left are not in the 3 times table.
Workbook Go further 6: Mr Lim has between 20 and 30 sweets. When he puts them in groups of 4, 3 are left over. When he puts them in groups of 5, 2 are left over. How many sweets does he have?	27. Groups of 4 with 3 left: 23 or 27. Groups of 5 with 2 left: 22 or 27. Only 27 is in both lists.
Workbook Go further 7: A pen has hens and goats. There are 2 more hens than goats. There are 22 legs altogether. How many hens and how many goats?	5 hens and 3 goats. Guess and check: 1 goat and 3 hens make 10 legs; 2 and 4 make 16; 3 and 5 make 22. Check: 12 + 10 = 22.

These are the questions on the two Think harder pages in the workbook (Think harder, then Go further). More are in the separate Question Bank.

Investigation (several answers are right)	Model answer and what to look for
Workbook Investigation (last core page): Where tables meet. (a) List the numbers up to 50 that are in BOTH the 2 times table and the 5 times table. What do you notice about them? (b) Now find the numbers up to 50 that are in both the 3 and 5 tables. (c) Is every number in both the 2 and 5 tables also in the 10 table? Explain. Easy first step: write the 5 times table up to 50, then circle the even numbers.	(a) 10, 20, 30, 40, 50: they jump by 10 and are the 10 times table. (b) 3 and 5: 15, 30, 45 (jump by 15). (Extension: 4 and 5 give 20, 40, jump by 20.) (c) Yes to both, brute-forced up to 100: a number in the 10 table is 10 x something = 2 x (5 x something) = 5 x (2 x something). Child-level reason: a number in the 2 table is even and a number in the 5 table ends in 0 or 5, so a number in both ends in 0, which is the 10 table; and every number in the 10 table ends in 0, so it is even and in the 5 table. Look for: organised lists (writing both tables out and circling), the noticing that the jumps are 10, 15, 20, and a test of “always”.

72–77 min Talk time

DO Bring the group back to a circle (or camera on, mics on). You choose volunteers first: "Who would like to share?" If nobody offers, ask the pairs to share what their partner said. Never cold-call a child by name. Any child may say "pass" at any time and that is always fine. Do not keep public tallies, scores or rankings (no counting who got most right, no "first to finish"). Invite the whole group to think about whether they would like a turn, never one child.

SAY *“Ravi shares stickers equally among 4 friends. Can the number left over ever be 4 or more?”*

“How did a table help you find how many hens and goats?”

Discussion question	Listen for

Ravi shares stickers equally among 4 friends. Can the number left over ever be 4 or more? Why?	No. If 4 or more were left, Ravi could give each friend one more, so the leftover is always 0, 1, 2 or 3. The table shows that every swap changes the legs by the same amount, so we can see how many swaps we need.

77–80 min Investigation share (3 minutes)

DO Show the Investigation slide. The Investigation is on the last core page of the workbook and is open to every child from question 3 on. Ask the whole group: “Who would like to share one way they tried it? Passing is fine.” Volunteers only, no names, no public tallies. Several answers are right: praise the method, not the speed. If nobody has started, give pairs one minute for the easy first step and set the rest as take-home.

Investigation	Model answer and what to look for
Investigation: Where tables meet	(a) 10, 20, 30, 40, 50: they jump by 10 and are the 10 times table. (b) 3 and 5: 15, 30, 45 (jump by 15). (Extension: 4 and 5 give 20, 40, jump by 20.) (c) Yes to both, brute-forced up to 100: a number in the 10 table is 10 x something = 2 x (5 x something) = 5 x (2 x something). Child-level reason: a number in the 2 table is even and a number in the 5 table ends in 0 or 5, so a number in both ends in 0, which is the 10 table; and every number in the 10 table ends in 0, so it is even and in the 5 table. Look for: organised lists (writing both tables out and circling), the noticing that the jumps are 10, 15, 20, and a test of “always”.

80–87 min Exit ticket

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide.

Exit ticket	Answer
A pen has hens and goats. There are 5 animals and 14 legs. How many hens and how many goats?	3 hens and 2 goats (6 + 8 = 14 legs).

87–89 min Looking back (2 minutes)

SAY *“Looking back takes 2 minutes. Find the small box called Looking back on your page. Finish one line, two lines or all three, and tick what helped you. You may pass on any line. This box is for you. If you tick "I am still stuck", your tutor will read it at the next lesson. Nothing is shared with the group.”*

DO Nothing is collected in front of the group and nobody has to read theirs out. The Looking back box is for the child. A paper box cannot be read live, so the tutor reads the journals at the next lesson, while stamping, and never by private message or photo. If a child ticked still stuck, start the next open session with a whole-group invitation (for example: "Anyone who would like another

look at last lesson's idea, stay with me and we will do it together"), with a parent nearby; never single out one child. A child may volunteer one sentence, never by name unless they choose to. No public tallies. Praise the strategy, not the speed.

89–90 min Closing and stamps

SAY *"Today you drew arrays, made tables and checked both clues. Next time a puzzle has two kinds of things mixed up, start with all of one kind and swap one at a time."*

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide (this is for you only, never shared with the children). If a child marked D or H, plan two minutes of catch-up at the start of the next lesson.

Think harder and Go further

Find all the arrays (at least 2 rows, at least 2 in each row) for 16 counters. (3 ways: 2 rows of 8, 4 rows of 4, 8 rows of 2.)

Answer key (workbook)

Q1 (a) 4 × 5 = 20 and 5 × 4 = 20. (b) 20 ÷ 4 = 5 and 20 ÷ 5 = 4.

Q2 (a) 30 cups (3 rows of 10). (b) 18 cups. We can see 12 (10 in the top row and 1 more in each of the other 2 rows), and 30 − 12 = 18. (c) e.g. "3 rows of 10 is 30, then take away the 12 I can see."

Q3 4 hens and 3 goats. Table: 7 hens 0 goats 14 legs; 6, 1, 16; 5, 2, 18; 4, 3, 20. Each swap adds 2 legs. Check: 8 + 12 = 20 legs and 4 + 3 = 7 animals.

Q4 (a) 5 stickers each (4 × 5 = 20). (b) 3 left over. (c) 1 more sticker: 24 stickers give 6 each with none left over.

Q5 (a) 6 vans. 5 vans hold 25 children and the other 3 children need a van too. (b) 2 empty seats (6 vans hold 30, and 30 − 28 = 2).

Q6 4 ways: 2 rows of 10, 4 rows of 5, 5 rows of 4, 10 rows of 2. 3 rows does not work (3 rows of 6 is 18, 3 rows of 7 is 21). More than 10 rows would leave fewer than 2 chairs in a row.

Story try-it (Shakuntala Devi). 7 × 5 = 35, 9 × 4 = 36, 6 × 10 = 60. Biggest: 6 × 10 = 60. Smallest: 7 × 5 = 35.

Investigation (Where tables meet). (a) 10, 20, 30, 40, 50: they jump by 10 and are the 10 times table. (b) 3 and 5: 15, 30, 45 (jump by 15). (Extension: 4 and 5 give 20, 40, jump by 20.) (c) Yes to both, brute-forced up to 100: a number in the 10 table is 10 x something = 2 x (5 x something) = 5 x (2 x something). Child-level reason: a number in the 2 table is even and a number in the 5 table ends in 0 or 5, so a number in both ends in 0, which is the 10 table; and every number in the 10 table ends in 0, so it is even and in the 5 table. Look for: organised lists (writing both tables out and circling), the noticing that the jumps are 10, 15, 20, and a test of “always”.

Lesson 4: Pattern Machines

Week 2 (second lesson) • Patterns with Multiplication • Workbook: Lesson 4

Objective	Find far places in a repeating pattern by counting in groups (the ends of the groups are in a times table); count how many of one shape there are; describe a growing pattern with a table and a rule and use it to go forwards and backwards; find the numbers that two counting patterns share.
Thinking skills	Look for a pattern, make a table, make a list, explain a rule
Materials	Linking cubes in 3 colours (about 20 per pair), mini whiteboards, matchsticks or lolly sticks (about 25 per pair), counters.
Online set-up	Children use coloured pencils and squared paper (or Lego bricks) for the cube train, and draw sticks for the growing pattern. The tutor shares the slides and the children show their tutor their tables on camera or keep them to share if they want.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0–1 min Opening

SAY *"Hello, pattern machines! Today we find out what is far, far along a pattern, like the 41st shape, without drawing all 41."*

"The secret is groups. If we know the group, the times tables tell us where every group ends."

DO Greet each child by name. Show the lesson title slide. Read "Today we will…" aloud.

1–4 min Sneak-peek puzzles (3 minutes)

SAY *"These are sneak-peek puzzles. They are ideas we have not learnt yet, so guessing is welcome and so is saying "I do not know". Nobody is marked and nothing is read out. Start with question 1: use objects, a drawing or counting. Then try question 2. Show your answer on your own paper to your tutor on camera, or keep it to share if you want. If you would like help, the tutor will invite the whole group to ask in the open session, and a parent can stay nearby."*

DO Show the Sneak-peek puzzles slide at minute 1 (3 minutes: about 1 minute for question 1, then 2 minutes for question 2). Say out loud: these are ideas we have not learnt yet; guessing and "I do not know" are welcome; nobody is marked.

Answers stay private. Each child holds their own paper to the camera for you only (look, do not read out), or writes it on their own paper and shows only you on camera. Never raise whiteboards together and never show a board in front of the group. Never ask a named child "how do you know?" Ask for volunteers: "Would anyone like to tell us how they started?"

Question 1 is an access question: every child can try it with objects, a drawing or counting, so everyone has a first win. Question 2 is the thinking question. Do not mark either one. Click to show the answers at the end as "one way to think about it", without saying who had what.

Use what you see only to plan. Invite the WHOLE group at once, never one child: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Never announce who is on which page and never name a child in front of the group. Children choose, and may move either way at any time. Pairs are not split unless both children agree. Everybody joins the explore, the mini-lesson, the check-together slides and talk time.

Sneak-peek puzzle	Answer and what to listen for
Question 1 (access: objects, drawing or counting): Draw this pattern up to the 9th shape: circle, circle, square, circle, circle, square ... What is the 9th shape?	A square (the squares are the 3rd, 6th and 9th shapes). Listen for: Drawing it out is a fine method. Listen for "every 3rd one is a square".
Question 2 (thinking): A pattern repeats: star, moon, moon, sun, star, moon, moon, sun ... What is the 22nd shape? How do you know?	A moon. Listen for: Groups of 4: 20 is the end of a group (5 groups of 4), so the 21st is a star and the 22nd is a moon. Drawing 22 shapes is also fine.

DO The neutral line “After question 3 you may choose: carry on, or try Think harder.” is printed on page 1 of the workbook, next to the lesson title. Nobody is labelled and nothing is announced.

4–8 min Warm-up

DO Pattern rhythm: clap, clap, stomp, clap, clap, stomp ... Keep it going. What do we do on beat 9? Beat 10? Beat 20?

SAY *“Clap, clap, stomp. Clap, clap, stomp. Keep it going with me!”*

“Freeze! What will we do on beat 9? On beat 10? How do you know?”

DO Keep the game to two minutes (4 to 6). Then the quick questions (6 to 8): ask questions 1, 3 and 5; questions 2 and 4 are spare. Children answer by thumbs, a whisper to a partner. Do not name a child.

Ask	Expect
4, 8, 12, __	16
50, 45, 40, __	35
3, 6, 9, 12, __	15
Count in 4s from 4. What is the 10th number?	40
2, 5, 8, 11, __	14

8–20 min Explore (in pairs, with objects)

SAY *“Our cube train goes red, red, blue, red, red, blue. What colour do you think cube 20 will be? Write your guess before you build.”*

DO 1. Build a cube train: red, red, blue, red, red, blue ...

2. Stop after 9 cubes.

3. Predict: what colour is cube 20? Write it down.

4. Build to check. Which cubes are blue?

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK Stuck: put a finger on each blue cube. Say their numbers. What do you notice?

The blue cubes are at 3, 6, 9 … Which table is that? Keep counting in 3s.

Finished early: what colour is cube 30? Cube 31? Explain without building.

DO Go further (a second what-if, so nobody waits): show the Explore: go further slide or post it in the chat when pairs go to their rooms. Any child may try it any time, as a choice. Never name who does. Question: what if the cube train is red, red, blue, green, red, red, blue, green …? What colour is cube 20? What colour is cube 30? Answer: Cube 20 is green (the green cubes are 4, 8, 12, 16, 20). Cube 30 is red (28 is green, 29 and 30 are red).

DO At about minute 14, say "Pencils down" and bring the group back together. Take the three questions below, one at a time (about 6 minutes).

Ask	Expect
What colour is cube 20?	Red.
How did you know without building?	The blue cubes are at 3, 6, 9 … 18, so 19 and 20 are red.
Which cubes are blue?	The 3 times table: 3, 6, 9, 12, 15, 18 …

20–32 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"Sun, moon, star, sun, moon, star. The group is 3 shapes long. Every group ends with a star, at 3, 6, 9, 12: the 3 times table. Place 14? 12 is the end of a group, so 13 starts again with a sun, and 14 is a moon."*

"Where is the 5th star? There is one star at the end of each group, so the 5th star ends the 5th group: 5 × 3 = 15."

"Squares made of sticks: 4, 7, 10. Each new square needs only 3 more sticks because it shares a side. I make a table and keep adding 3."

Worked example on the slides	Answer
Far places: count in groups. Sun, moon, star, sun, moon, star … The group is 3 shapes. Groups end at 3, 6, 9, 12. What is in place 14?	A moon (12 ends a group; 13 sun, 14 moon).

Where is the 5th star?. One star in each group, at the end. Stars are at 3, 6, 9, 12, 15. Where is the 5th star? How many moons are in the first 21 shapes?	Place 15. 7 moons (21 shapes is 7 groups).

SAY *"The key idea today is: count in groups. Find the group size. The groups end at the times-table numbers. Jump to the end of a group, then count on. Say it with me."*

DO Show the key idea slide at about minutes 31 to 32. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Saying that a place in the times table (like 20 for a group of 5) is the FIRST shape of a group, when it is the LAST shape.
- Counting the whole pattern one by one to the far place instead of jumping to the end of a group.
- Growing patterns: multiplying by the step (10 squares × 3 = 30) and forgetting the start (the extra stick), instead of using the table or the rule.

32–36 min Brain break

DO Pattern moves: jump, jump, squat, jump, jump, squat … 15 moves. What is move 15?

Water. Toilet break for anyone who needs it. Keep it to four minutes.

DO Story break (optional, 2 minutes): do the movement game for 2 minutes (32 to 34), then show the "Story break: Hertha Ayrton" slide (34 to 36). Read the story aloud (about 30 seconds): "Hertha Ayrton was born in England in 1854. She was a scientist and an inventor. She studied the patterns of ripples that water makes in sand. She built a model sea in a zinc bath to study them. She saw that the ripples came in matching pairs, one on each side of the first ridge. In 1904 she was the first woman to present her own paper at the Royal Society, a famous group of scientists."

DO Then ask: "Would anyone like to try this one?" Try it: Ripples on a beach repeat: ridge, dip, dip, dip, then ridge, dip, dip, dip again. Is ripple number 20 a ridge or a dip? What about ripple number 21? Volunteers only, never name a child, nobody is marked, passing is fine. SKIP the story break, without comment, if you are behind time or the group needs the full four minutes.

36–72 min Worksheet (pairs, then alone, then stretch)

DO Minute 36: give out the workbook. Read the title of Lesson 4 aloud. Say "Pairs first. Talk before you write."

Minutes 36 to 50: pairs work together on the first page (online, visit every shared breakout room by minute 47 so no pair works unseen for long; in Lessons 5, 6 and 10 a short teach slide comes first, 36 to 38). Minutes 50 to 52: check page 1 together. Minutes 52 to 57: short teaching for page 2. Minutes 57 to 69: each child works alone on the second page. Minutes 69 to 72: check page 2 together. Think harder, Go further and the Investigation are open to any child at any time after question 3.

SAY *"Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking."*

DO Circulate and ask:

- Ask: "How long is one group? Where does each group end?"
- Ask: "Is that place the first shape of a group or the last?"
- Ask: "What is the rule? Does it work for every pair in your table?"

IF STUCK Hint 1: "Draw a line after each group. Count the groups."

Hint 2: "Find the nearest times-table number before your place. That shape ends a group."

Hint 3: "Make a table and look at how much it goes up each time."

DO A choice, not a track: after question 3 any child may carry on or try Think harder (the neutral line is printed on page 1). Invite the WHOLE group at once, never one child: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Never announce who is on which page, never name a child in front of the group, and do not split a pair unless both children agree. Children may move either way at any time. The Investigation is printed on the last core page, so every child sees it.

Think harder and Go further questions

Question	Answer
Workbook Think harder 1: Look at the shape pattern in the Start here page again (places 1 to 10). How many circles are there in the first 30 shapes? How do you know?	12 circles (30 shapes is 6 groups of 5, with 2 circles in each group).
Workbook Think harder 2: A pattern goes A, B, B, C, A, B, B, C ... What is the 25th letter?	A. 24 ends a group of 4 ($4 \times 6 = 24$), so 25 starts again with A.
Workbook Think harder 3: Beads go red, yellow, yellow, red, yellow, yellow ... How many yellow beads are in the first 18 beads?	12 yellow beads (18 beads is 6 groups of 3, with 2 yellow in each).
Workbook Think harder 4: Houses are made of sticks in a row. 1 house uses 5 sticks, 2 houses use 9 sticks, 3 houses use 13 sticks. How many sticks are needed for 6 houses?	25 sticks. The rule is add 4: 5, 9, 13, 17, 21, 25.
Workbook Go further 5: Ali counts in 3s from 3. Mei counts in 4s from 4. They both say 12. Predict the next number they will BOTH say. Then list every number up to 100 that they both say. What do you notice? Why?	24 is next. They both say 12, 24, 36, 48, 60, 72, 84 and 96 (8 numbers): the numbers go up by 12 each time, because 12 is the first number in both the 3 and 4 times tables, and each next one is 12 more.
Workbook Go further 6: Siti builds towers of cubes. Tower 1 has 2 cubes, tower 2 has 5 cubes, tower 3 has 8 cubes. The towers keep growing in the same way. Which tower has 29 cubes? Is there a tower with exactly 50 cubes? With exactly 100 cubes? Explain how you know.	Tower 10 has 29 cubes (the rule is add 3: 2, 5, 8, 11, ... 29). Tower 17 has 50 cubes. No tower has 100: every tower has 1 less than a number in the 3 times table (tower 10: $30 - 1 = 29$), and 101 is not in the 3 times table (99 and 102 are).

Workbook Go further 7: Frog A starts at 0 and jumps 4 each time: 4, 8, 12 … Frog B starts at 2 and jumps 3 each time: 5, 8, 11 … Which numbers up to 30 do both frogs land on?	8 and 20. Frog A: 4, 8, 12, 16, 20, 24, 28. Frog B: 5, 8, 11, 14, 17, 20, 23, 26, 29.

These are the questions on the two Think harder pages in the workbook (Think harder, then Go further). More are in the separate Question Bank.

Investigation (several answers are right)	Model answer and what to look for
Workbook Investigation (last core page): Three counting patterns. Pattern A: 1, 4, 7, 10 … (add 3 each time). Pattern B: 2, 5, 8, 11 … Pattern C: 3, 6, 9, 12 … (a) Write each pattern up to 30. (b) Pick any number from 1 to 30. Which pattern is it in? Can a number be in two patterns? Can a number be in none? (c) Why does this happen? Easy first step: write pattern A, 1, 4, 7, 10 …, up to 30.	(a) A: 1, 4, 7, 10, 13, 16, 19, 22, 25, 28. B: 2, 5, 8, 11, 14, 17, 20, 23, 26, 29. C: 3, 6, 9, 12, 15, 18, 21, 24, 27, 30. (b) Every number is in exactly ONE pattern (brute-forced for 1 to 99). (c) Counting in 3s, a number is either on the 3s (pattern C), 1 more than a number on the 3s (A), or 2 more (B); there is no other choice. Extension (not printed): if each pattern added 4 you would need 4 patterns, starting at 1, 2, 3 and 4. Look for: a complete listing, the claim “exactly one” tested on more than one number, and a reason that uses the 3 times table.

72–77 min Talk time

DO Bring the group back to a circle (or camera on, mics on). You choose volunteers first: "Who would like to share?" If nobody offers, ask the pairs to share what their partner said. Never cold-call a child by name. Any child may say "pass" at any time and that is always fine. Do not keep public tallies, scores or rankings (no counting who got most right, no "first to finish"). Invite the whole group to think about whether they would like a turn, never one child.

SAY *“A pattern repeats every 5 shapes. Is the 10th shape always the same as the 20th?”*

“What if the pattern repeats every 3 shapes?”

Discussion question	Listen for
A pattern repeats every 5 shapes. Is the 10th shape always the same as the 20th? What if the pattern repeats every 3 shapes?	Every 5: yes, 10 and 20 are both in the 5 times table, so both are the last shape of a group. Every 3: no. 9 ends a group, so 10 is the 1st shape; 18 ends a group, so 20 is the 2nd shape.

77–80 min Investigation share (3 minutes)

DO Show the Investigation slide. The Investigation is on the last core page of the workbook and is open to every child from question 3 on. Ask the whole group: “Who would like to share one way they tried it? Passing is fine.” Volunteers only, no names, no public tallies. Several answers are right: praise the method, not the speed. If nobody has started, give pairs one minute for the easy first step and set the rest as take-home.

Investigation	Model answer and what to look for
Investigation: Three counting patterns	(a) A: 1, 4, 7, 10, 13, 16, 19, 22, 25, 28. B: 2, 5, 8, 11, 14, 17, 20, 23, 26, 29. C: 3, 6, 9, 12, 15, 18, 21, 24, 27, 30. (b) Every number is in exactly ONE pattern (brute-forced for 1 to 99). (c) Counting in 3s, a number is either on the 3s (pattern C), 1 more than a number on the 3s (A), or 2 more (B); there is no other choice. Extension (not printed): if each pattern added 4 you would need 4 patterns, starting at 1, 2, 3 and 4. Look for: a complete listing, the claim “exactly one” tested on more than one number, and a reason that uses the 3 times table.

80–87 min Exit ticket

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide.

Exit ticket	Answer
The pattern goes triangle, circle, square, triangle, circle, square … What is the 17th shape? How do you know?	A circle. 15 ends a group (5 groups of 3), so 16 is a triangle and 17 is a circle.

87–89 min Looking back (2 minutes)

SAY *“Looking back takes 2 minutes. Find the small box called Looking back on your page. Finish one line, two lines or all three, and tick what helped you. You may pass on any line. This box is for you. If you tick "I am still stuck", your tutor will read it at the next lesson. Nothing is shared with the group.”*

DO Nothing is collected in front of the group and nobody has to read theirs out. The Looking back box is for the child. A paper box cannot be read live, so the tutor reads the journals at the next lesson, while stamping, and never by private message or photo. If a child ticked still stuck, start the next open session with a whole-group invitation (for example: "Anyone who would like another look at last lesson's idea, stay with me and we will do it together"), with a parent nearby; never single out one child. A child may volunteer one sentence, never by name unless they choose to. No public tallies. Praise the strategy, not the speed.

89–90 min Closing and stamps

SAY *“You did not draw 41 shapes today. You found the group, used a times table and counted on. That is how pattern machines think.”*

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide (this is for you only, never shared with the children). If a child marked D or H, plan two minutes of catch-up at the start of the next lesson.

Think harder and Go further

Make your own repeating pattern of 4 shapes. Write 3 "far place" questions about it for a partner, with the answers.

Answer key (workbook)

Q1 (a) Square (20 ends a group of 5). (b) Circle (21 triangle, 22 circle, 23 circle). (c) Triangle: 40 ends a group (5 × 8 = 40), so 41 starts again.

Q2 (a) A (16 ends a group of 4; 17 is M, 18 is A). (b) Place 40 (every H ends a group: 4, 8, 12 … the 10th is 4 × 10). (c) 7 Ts. They are in places 3, 7, 11, 15, 19, 23, 27; 28 ends the 7th group, and places 29 and 30 are M and A.

Q3 (a) 18 yellow beads (12 red is 6 groups of 2 red; each group has 3 yellow). (b) 30 beads (6 groups of 5).

Q4 (a) 5, 10, 15, 20, 25, 30. (b) 50 petals. (c) 7 flowers. (d) 4 flowers, with 3 petals left over.

Q5 (a) 4, 6, 8, 10, 12, 14. (b) 22 children. (c) 9 tables. (d) Each new table adds 1 seat on the top and 1 on the bottom; the 2 seats at the ends stay the same.

Q6 Ali (12 numbers, ending at 36): 3, 6, 9, 12, 15, 18, 21, 24, 27, 30, 33, 36. Mei (9 numbers, ending at 36): 4, 8, 12, 16, 20, 24, 28, 32, 36. Both: 12, 24, 36. They go up by 12 each time.

Story try-it (Hertha Ayrton). The pattern repeats in groups of 4. Ripple 20 is the last of the fifth group (5 × 4 = 20), so it is a dip. Ripple 21 starts a new group, so it is a ridge.

Investigation (Three counting patterns). (a) A: 1, 4, 7, 10, 13, 16, 19, 22, 25, 28. B: 2, 5, 8, 11, 14, 17, 20, 23, 26, 29. C: 3, 6, 9, 12, 15, 18, 21, 24, 27, 30. (b) Every number is in exactly ONE pattern (brute-forced for 1 to 99). (c) Counting in 3s, a number is either on the 3s (pattern C), 1 more than a number on the 3s (A), or 2 more (B); there is no other choice. Extension (not printed): if each pattern added 4 you would need 4 patterns, starting at 1, 2, 3 and 4. Look for: a complete listing, the claim “exactly one” tested on more than one number, and a reason that uses the 3 times table.

Lesson 5: Mystery Number Machines

Week 3 (first lesson) • Working Backwards • Workbook: Lesson 5

Objective	Find a mystery number in two-step puzzles with +, −, × and ÷ by working backwards (undo the last step first with the opposite operation), check by going forwards, and use before-and-after bar models for stories where an amount changes.
Thinking skills	Inverse operations, working backwards, guess and check, before-and-after bar models, checking forwards
Materials	Mini whiteboards, counters, paper cups, number machine cards (× 2, × 3, × 4, × 5, × 10 and + or − cards from 1 to 20), paper strips for bars.
Online set-up	Machine cards shown on the slide; children write on whiteboards or paper and show their tutor on camera or keep it to share if they want. For bars, children draw on paper or fold paper strips into equal parts.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0–1 min Opening

SAY *"Good morning, inquirers! Today you are code-breakers. A number went into a secret machine and something came out. Your job is to run the machine backwards and find the number that went in."*

"Remember our rule: notice it, test it, explain why."

DO Greet each child by name. Show the lesson title slide. Read "Today we will..." aloud.

1–4 min Sneak-peek puzzles (3 minutes)

SAY *"These are sneak-peek puzzles. They are ideas we have not learnt yet, so guessing is welcome and so is saying "I do not know". Nobody is marked and nothing is read out. Start with question 1: use objects, a drawing or counting. Then try question 2. Show your answer on your own paper to your tutor on camera, or keep it to share if you want. If you would like help, the tutor will invite the whole group to ask in the open session, and a parent can stay nearby."*

DO Show the Sneak-peek puzzles slide at minute 1 (3 minutes: about 1 minute for question 1, then 2 minutes for question 2). Say out loud: these are ideas we have not learnt yet; guessing and "I do not know" are welcome; nobody is marked.

Answers stay private. Each child holds their own paper to the camera for you only (look, do not read out), or writes it on their own paper and shows only you on camera. Never raise whiteboards together and never show a board in front of the group. Never ask a named child "how do you know?" Ask for volunteers: "Would anyone like to tell us how they started?"

Question 1 is an access question: every child can try it with objects, a drawing or counting, so everyone has a first win. Question 2 is the thinking question. Do not mark either one. Click to show

the answers at the end as "one way to think about it", without saying who had what.

Use what you see only to plan. Invite the WHOLE group at once, never one child: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Never announce who is on which page and never name a child in front of the group. Children choose, and may move either way at any time. Pairs are not split unless both children agree. Everybody joins the explore, the mini-lesson, the check-together slides and talk time.

Sneak-peek puzzle	Answer and what to listen for
Question 1 (access: objects, drawing or counting): A machine adds 7. The OUT is 12. What was the IN? (Count back with counters if you like.)	5 (5 + 7 = 12). Listen for: IN is what goes in and OUT is what comes out. Undo = do the opposite: 12 - 7. This is a first win for everyone.
Question 2 (thinking): A machine doubles, then adds 7. The OUT is 17. What was the IN? How do you know?	5 (5 doubled is 10, and 10 + 7 = 17). Listen for: Undo the last step first: 17 - 7 = 10, then half of 10 is 5. A guess that was tested (5 x 2 + 7 = 17) is just as good.

DO The neutral line “After question 3 you may choose: carry on, or try Think harder.” is printed on page 1 of the workbook, next to the lesson title. Nobody is labelled and nothing is announced.

4–8 min Warm-up

DO Guess my machine. The tutor is a secret one-step machine. Call out an IN number, hear the OUT, and guess the rule. Then the tutor gives only an OUT: what went IN?

SAY *“I am a machine. Give me a number and I will tell you what comes out. Ready? You say 4, I say 12. You say 10, I say 30. What am I doing?”*

“Now I only tell you what comes out: 18. What went in? How did you work it out?”

DO Keep the game to two minutes (4 to 6). Then the quick questions (6 to 8): ask questions 1, 3 and 5; questions 2 and 4 are spare. Children answer by thumbs, a whisper to a partner. Do not name a child.

Ask	Expect
I think of a number. I multiply it by 2. I get 18. What is it?	9
I think of a number. I divide it by 5. I get 4. What is it?	20
I think of a number. I take away 25. I get 40. What is it?	65
What is the opposite of × 4?	÷ 4
I think of a number. I multiply it by 3, then add 4. I get 19. What is it?	5

8–20 min Explore (in pairs, with objects)

SAY *"Here are two machine cards: × 2 and + 6. If I put 4 in, what comes out? Now I swap the cards: + 6 first, then × 2. Will the same number come out? Thumbs up or down before we test it."*

DO 1. Pick one × card and one + or − card. Put them in a row: that is your machine.

2. Secretly choose a start number. Run it through both cards. Say only the OUT.

3. Your partner works backwards to find your start number. Check it together.

4. Swap the order of the two cards. Put the same start number in. Is the OUT the same?

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK Stuck: write the machine as boxes and arrows, then put your partner's OUT number in the last box.

Mixed up: "What was the LAST thing the machine did? Undo that first."

Finished early: find an OUT number your partner can NOT undo with whole numbers. Why does it not work?

DO Go further (a second what-if, so nobody waits): show the Explore: go further slide or post it in the chat when pairs go to their rooms. Any child may try it any time, as a choice. Never name who does. Question: what if the machine adds 7 FIRST and then doubles? Put IN 5 into it. Is the OUT the same as before (17)? Answer: No: 5 + 7 = 12, doubled is 24. The order of the steps matters.

DO At about minute 14, say "Pencils down" and bring the group back together. Take the three questions below, one at a time (about 6 minutes).

Ask	Expect
How did you find your partner's start number?	Undo the last card first with its opposite, then undo the first card.
What is the opposite of × 3? Of − 8?	÷ 3. + 8.
Does it matter which card goes first?	Yes. Start 4: × 2 then + 6 gives 14, but + 6 then × 2 gives 20.

20–32 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"Mystery number × 3, then + 4, gives 25. I go backwards. The last step was + 4, so I undo it first with − 4: 25 − 4 = 21. The step before was × 3, so I undo it with ÷ 3: 21 ÷ 3 = 7. Check forwards: 7 × 3 = 21, 21 + 4 = 25. ✓"*

"Ben gave away 18 stamps, then bought 25, and now has 60. I draw the bars and start at the end. Before he bought 25 he had 60 − 25 = 35. Before he gave away 18 he had 35 + 18 = 53. Check: 53 − 18 = 35, 35 + 25 = 60. ✓"

"Mei gave away half her beads, then used 5. She has 9 left. The half she kept is 5 + 9 = 14. The other half is also 14. So she had 14 + 14 = 28."

Worked example on the slides	Answer
Two steps: undo the last step first. ? → × 3 → + 4 → 25 Undo + 4: 25 − 4 = 21. Undo × 3: 21 ÷ 3 = ? Check forwards.	7 (7 × 3 = 21, 21 + 4 = 25).
Before and after: draw the bars. Ben had some stamps. He gave 18 to Mei. Then he bought 25 more. Now he has 60. Before he bought 25: 60 − 25 = 35. Before he gave 18: 35 + 18 = ?	53 stamps (check 53 − 18 = 35, 35 + 25 = 60).

SAY *"The key idea today is: undo it backwards. Start at the end. Undo the LAST step first with its opposite, then the step before. Check by going forwards. Say it with me."*

DO Show the key idea slide at about minutes 31 to 32. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Doing the same step again instead of the opposite (27 × 2 instead of 27 ÷ 2).
- Undoing the steps in the wrong order: the first step first instead of the last step first.
- Bar stories: stopping at one part (one bag or one half) instead of finding the whole, or forgetting to undo one of the changes.

32–36 min Brain break

DO Machine moves. The tutor calls a machine ("× 2 jumps!", "+ 3 claps!") on a number on the board, and everyone does the OUT number of moves.

Water. Toilet break for anyone who needs it. Keep it to four minutes.

DO Story break (optional, 2 minutes): do the movement game for 2 minutes (32 to 34), then show the "Story break: Mary Somerville" slide (34 to 36). Read the story aloud (about 30 seconds): "Mary Somerville grew up in Scotland about 240 years ago. As a girl, she often told how she found a puzzle page in a magazine. It was full of strange letters, mostly X and Y. A friend said it was a kind of arithmetic called algebra, but could not explain it. Mary wanted to know more, so she found books and taught herself. In 1831 she published a book that explained a hard book about the stars." Say clearly: From her own memories, told in later accounts. The exact age is not certain.

DO Then ask: "Would anyone like to try this one?" Try it: Mary's X was a mystery number. I double X, then take away 3, and the answer is 9. Work backwards to find X. Volunteers only, never name a child, nobody is marked, passing is fine. SKIP the story break, without comment, if you are behind time or the group needs the full four minutes.

36–72 min Worksheet (pairs, then alone, then stretch)

DO Minute 36: give out the workbook. Read the title of Lesson 5 aloud. Say "Pairs first. Talk before you write."

Minutes 36 to 50: pairs work together on the first page (online, visit every shared breakout room by minute 47 so no pair works unseen for long; in Lessons 5, 6 and 10 a short teach slide comes first, 36 to 38). Minutes 50 to 52: check page 1 together. Minutes 52 to 57: short teaching for page 2.

Minutes 57 to 69: each child works alone on the second page. Minutes 69 to 72: check page 2 together. Think harder, Go further and the Investigation are open to any child at any time after question 3.

SAY *"Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking."*

DO Circulate and ask:

- Ask: "What was the last step? What is its opposite?"
- Ask: "Show me your check going forwards. Do you get the OUT number?"
- Ask: "Point to your bar. Which part do you know? Which part is the mystery?"

IF STUCK Hint 1: "Draw the machine boxes. Write the OUT in the last box."

Hint 2: "Cover the first step with your finger. Undo only the last step."

Hint 3: "For bars: start from what is left now and put back each change, from the end to the start."

DO A choice, not a track: after question 3 any child may carry on or try Think harder (the neutral line is printed on page 1). Invite the WHOLE group at once, never one child: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Never announce who is on which page, never name a child in front of the group, and do not split a pair unless both children agree. Children may move either way at any time. The Investigation is printed on the last core page, so every child sees it.

Think harder and Go further questions

Question	Answer
Workbook Think harder 1: A machine multiplies by a number, then adds a number. IN 2 gives OUT 11 and IN 5 gives OUT 20. Find what the machine does (try × 2, × 3, × 4 … with IN 2, then test IN 5). Then: what OUT does IN 10 give? Which IN gives OUT 26?	× 3, then + 5. IN 10 gives OUT 35. OUT 26 comes from IN 7 (26 − 5 = 21, 21 ÷ 3 = 7).
Workbook Think harder 2: Ben had some money. He spent $15 on a book. Then his dad gave him $10. Now he has $42. How much money did Ben have at first?	$47 (42 − 10 = 32, 32 + 15 = 47)
Workbook Think harder 3: Siti had some beads. She gave half of them to Mei. Then she bought 12 more. Now she has 30 beads. How many beads did Siti have at first?	36 (30 − 12 = 18 is half, 18 + 18 = 36)

Workbook Think harder 4: Ali, Ben and Siti have 40 marbles altogether. Ben has 5 more marbles than Ali. Siti has 8 more marbles than Ali. How many marbles does each child have? Draw a bar for each child.	Ali 9, Ben 14, Siti 17 (40 − 5 − 8 = 27, 27 ÷ 3 = 9; check 9 + 14 + 17 = 40)
Workbook Go further 5: Siti had some sweets. She gave Ben 1 more than half of them. Now she has 7 sweets left. How many sweets did she have at first?	16 (7 + 1 = 8 is half, 8 + 8 = 16; she gave Ben 9)
Workbook Go further 6: A machine does × 2, then − 1. Mei puts a number through the machine 3 times in a row (each OUT goes back in). The final OUT is 17. What number did she start with?	3 (17 + 1 = 18, 18 ÷ 2 = 9; 9 + 1 = 10, 10 ÷ 2 = 5; 5 + 1 = 6, 6 ÷ 2 = 3; check 3 → 5 → 9 → 17)
Workbook Go further 7: Three boxes have 30 pens altogether. 5 pens are moved from box A to box B. Then 3 pens are moved from box B to box C. Now all three boxes have the same number of pens. How many pens were in each box at first?	A 15, B 8, C 7 (each ends with 10; undo the moves from the end)

These are the questions on the two Think harder pages in the workbook (Think harder, then Go further). More are in the separate Question Bank.

Investigation (several answers are right)	Model answer and what to look for
Workbook Investigation (last core page): A magic trick. Think of a number. Add 5. Double it. Take away 10. Halve it. (a) Try 5 different starting numbers. What do you notice about your final answer? (b) Is it ALWAYS true? Try a big number and a small number, then say why. (c) Make your own trick that always gives you back the number you started with. Test it on 3 numbers. Easy first step: do the trick with 3, then with 10, and write down each step.	(a) The final answer is always the number you started with (brute-forced for 1 to 999). (b) Adding 5 then doubling adds 10 and doubles the start; taking away 10 and halving undoes both: (n + 5) x 2 − 10 = n x 2, and half of that is n. A child reason: “double and halve undo each other, and add 5 doubled is 10, which I take away”. (c) Any valid trick, for example: add 3, double, take away 6, halve; or times 4, add 8, divide by 4, take away 2. Check by testing three numbers. Look for: testing small and big numbers, a reason in words, and a trick that really works.

72–77 min Talk time

DO Bring the group back to a circle (or camera on, mics on). You choose volunteers first: "Who would like to share?" If nobody offers, ask the pairs to share what their partner said. Never cold-call a child by name. Any child may say "pass" at any time and that is always fine. Do not keep public tallies, scores or rankings (no counting who got most right, no "first to finish"). Invite the whole group to think about whether they would like a turn, never one child.

SAY *“To undo × 2, then + 6, with OUT 20, Mei takes away 6 first. Ali divides by 2 first. Who is right? Test both.”*

“Why do we undo the LAST step first?”

Discussion question	Listen for
To undo “× 2, then + 6” with OUT 20, Mei takes away 6 first. Ali divides by 2 first. Who is right?	Mei is right: 20 − 6 = 14, 14 ÷ 2 = 7, and 7 × 2 + 6 = 20. Ali gets 20 ÷ 2 = 10, 10 − 6 = 4, but 4 × 2 + 6 = 14, not 20. The last step happened last, so it must be undone first, like taking off shoes before socks.

77–80 min Investigation share (3 minutes)

DO Show the Investigation slide. The Investigation is on the last core page of the workbook and is open to every child from question 3 on. Ask the whole group: “Who would like to share one way they tried it? Passing is fine.” Volunteers only, no names, no public tallies. Several answers are right: praise the method, not the speed. If nobody has started, give pairs one minute for the easy first step and set the rest as take-home.

Investigation	Model answer and what to look for
Investigation: A magic trick	(a) The final answer is always the number you started with (brute-forced for 1 to 999). (b) Adding 5 then doubling adds 10 and doubles the start; taking away 10 and halving undoes both: (n + 5) x 2 − 10 = n x 2, and half of that is n. A child reason: “double and halve undo each other, and add 5 doubled is 10, which I take away”. (c) Any valid trick, for example: add 3, double, take away 6, halve; or times 4, add 8, divide by 4, take away 2. Check by testing three numbers. Look for: testing small and big numbers, a reason in words, and a trick that really works.

80–87 min Exit ticket

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide.

Exit ticket	Answer
I think of a number. I multiply it by 4, then take away 6. I get 14. What is my number?	5 (14 + 6 = 20, 20 ÷ 4 = 5; check 5 × 4 = 20, 20 − 6 = 14)

87–89 min Looking back (2 minutes)

SAY *“Looking back takes 2 minutes. Find the small box called Looking back on your page. Finish one line, two lines or all three, and tick what helped you. You may pass on any line. This box is for you. If you tick "I am still stuck", your tutor will read it at the next lesson. Nothing is shared with the group.”*

DO Nothing is collected in front of the group and nobody has to read theirs out. The Looking back box is for the child. A paper box cannot be read live, so the tutor reads the journals at the next lesson, while stamping, and never by private message or photo. If a child ticked still stuck, start the next open session with a whole-group invitation (for example: "Anyone who would like another

look at last lesson's idea, stay with me and we will do it together"), with a parent nearby; never single out one child. A child may volunteer one sentence, never by name unless they choose to. No public tallies. Praise the strategy, not the speed.

89–90 min Closing and stamps

SAY *"Today you ran machines backwards and drew before-and-after bars. Code-breakers always check forwards, and you did. Well done, inquirers!"*

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide (this is for you only, never shared with the children). If a child marked D or H, plan two minutes of catch-up at the start of the next lesson.

Think harder and Go further

Make a three-step machine with a × card, a ÷ card and a + card. Give a partner only the OUT. Can they find the IN? Which OUT numbers make your machine impossible to undo?

Answer key (workbook)

Q1 (a) 10 (b) 16 (c) 24 (d) 9 (undo the last step first: e.g. (a) 27 − 7 = 20, 20 ÷ 2 = 10)

Q2 (a) IN 4 → OUT 14; IN 10 → OUT 32; OUT 17 ← IN 5; OUT 29 ← IN 9. (b) IN 3, 5, 10, 20 give OUT 7, 11, 21, 41.

Q3 59 stickers (50 − 15 = 35, 35 + 24 = 59; check 59 − 24 = 35, 35 + 15 = 50)

Q4 30 sweets (one bag: 6 + 4 = 10; 3 bags: 10 × 3 = 30)

Q5 (a) Mei 18 (b) Ali 14 (c) No: the order of the steps matters (d) 7 (18 − 4 = 14, 14 ÷ 2 = 7)

Story try-it (Mary Somerville). X = 6. Undo −3: 9 + 3 = 12. Undo doubling: 12 ÷ 2 = 6. Check: 6 × 2 = 12 and 12 − 3 = 9.

Investigation (A magic trick). (a) The final answer is always the number you started with (brute-forced for 1 to 999). (b) Adding 5 then doubling adds 10 and doubles the start; taking away 10 and halving undoes both: (n + 5) x 2 − 10 = n x 2, and half of that is n. A child reason: "double and halve undo each other, and add 5 doubled is 10, which I take away". (c) Any valid trick, for example: add 3, double, take away 6, halve; or times 4, add 8, divide by 4, take away 2. Check by testing three numbers. Look for: testing small and big numbers, a reason in words, and a trick that really works.

Lesson 6: Market Day

Week 3 (second lesson) • Money • Workbook: Lesson 6

Objective	Read and write money in dollars and cents, pay exactly with the fewest notes and coins, make a list in order to find every way to pay, give change by counting on, decide whether there is enough money, and work backwards to find how much someone had at first.
Thinking skills	Systematic listing (make a list in order), working backwards, counting on, comparing, justifying "there are no more ways"
Materials	Play money: coins 5¢, 10¢, 20¢, 50¢, $1 and notes $2, $5, $10, $50 (about 6 of each per pair), price tags, mini whiteboards.
Online set-up	Print the cut-out coins and notes, or use real money at home. No play money? Draw each coin and note on paper. The shop and the toy stall are on the slides; children show their money to the camera.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0–1 min Opening

SAY *"Welcome to Market Day, inquirers! Today you are shoppers AND shopkeepers. You will pay exactly, give change, and find every way to pay without missing one."*

"Notice it, test it with play money, and explain why."

DO Greet each child by name. Show the lesson title slide. Read "Today we will..." aloud.

1–4 min Sneak-peek puzzles (3 minutes)

SAY *"These are sneak-peek puzzles. They are ideas we have not learnt yet, so guessing is welcome and so is saying "I do not know". Nobody is marked and nothing is read out. Start with question 1: use objects, a drawing or counting. Then try question 2. Show your answer on your own paper to your tutor on camera, or keep it to share if you want. If you would like help, the tutor will invite the whole group to ask in the open session, and a parent can stay nearby."*

DO Show the Sneak-peek puzzles slide at minute 1 (3 minutes: about 1 minute for question 1, then 2 minutes for question 2). Say out loud: these are ideas we have not learnt yet; guessing and "I do not know" are welcome; nobody is marked.

Answers stay private. Each child holds their own paper to the camera for you only (look, do not read out), or writes it on their own paper and shows only you on camera. Never raise whiteboards together and never show a board in front of the group. Never ask a named child "how do you know?" Ask for volunteers: "Would anyone like to tell us how they started?"

Question 1 is an access question: every child can try it with objects, a drawing or counting, so everyone has a first win. Question 2 is the thinking question. Do not mark either one. Click to show the answers at the end as "one way to think about it", without saying who had what.

Use what you see only to plan. Invite the WHOLE group at once, never one child: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Never announce who is on which page and never name a child in front of the group. Children choose, and may move either way at any time. Pairs are not split unless both children agree. Everybody joins the explore, the mini-lesson, the check-together slides and talk time.

Sneak-peek puzzle	Answer and what to listen for
Question 1 (access: objects, drawing or counting): Mei has one 50¢ coin, one 20¢ coin and one 10¢ coin. How much money is that? Use play coins or draw them.	80¢ (50 + 20 + 10). Listen for: Counting on from the biggest coin. This is a first win for everyone.
Question 2 (thinking): Mei has $5. Three drinks cost $1.60 each. Does she have enough? How much money is left?	Yes. 3 drinks cost $4.80, so 20¢ is left. Listen for: Listen for adding $1.60 three times, or $1 + $1 + $1 and then 60¢ three times (180¢ = $1.80). Counting up from $4.80 to $5 is a good method.

DO The neutral line “After question 3 you may choose: carry on, or try Think harder.” is printed on page 1 of the workbook, next to the lesson title. Nobody is labelled and nothing is announced.

4–8 min Warm-up

DO Market Day shop. Six items with price tags from 85¢ to $57.20. Pay exactly with play notes and coins. Who can pay with the fewest pieces?

SAY *“Here is our market. The sticker book costs $3.45. That means 3 dollars and 45 cents. Pay exactly with your play money.”*

“Now count your pieces. Who used the fewest? How did you choose?”

DO Keep the game to two minutes (4 to 6). Then the quick questions (6 to 8): ask questions 1, 3 and 5; questions 2 and 4 are spare. Children answer by thumbs, a whisper to a partner. Do not name a child.

Ask	Expect
Write 345¢ in dollars and cents.	$3.45
How many cents is $2.05?	205¢
Which is more: $1.50 or 105¢?	$1.50 (150¢)
Pay 75¢ with the fewest coins.	50¢ + 20¢ + 5¢
You have 60¢. A sticker costs 85¢. How much more do you need?	25¢

8–20 min Explore (in pairs, with objects)

SAY *"The toy stall has a kite, a ball, a yo-yo, bubbles and a sticker. You have $10. How many different baskets do you think cost EXACTLY $10? Write a guess."*

DO 1. You have $10. Look at the toy stall: kite $6, ball $4, yo-yo $3, bubbles $2, sticker $1.

2. Choose some toys (at most one of each). How much do they cost? How much change do you get?

3. Find every basket that costs EXACTLY $10.

4. Make a list in order so you do not miss one or count one twice.

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK Stuck: start with the most expensive toy. Put it in the basket. How much is left to spend?

Repeating baskets: "Kite and ball is the same basket as ball and kite."

Finished early: which basket gives you the most toys for $10? Which gives exactly $1 change?

DO Go further (a second what-if, so nobody waits): show the Explore: go further slide or post it in the chat when pairs go to their rooms. Any child may try it any time, as a choice. Never name who does. Question: what if the toy stall also sells a robot for $5? Now how many different baskets cost exactly $10? (At most one of each toy.) Answer: 5 baskets: kite + ball; kite + yo-yo + sticker; ball + sticker + robot; yo-yo + bubbles + robot; ball + yo-yo + bubbles + sticker.

DO At about minute 14, say "Pencils down" and bring the group back together. Take the three questions below, one at a time (about 6 minutes).

Ask	Expect
How many baskets cost exactly $10?	3: kite + ball; kite + yo-yo + sticker; ball + yo-yo + bubbles + sticker.
How do you know you have them all?	Start with the kite in the basket, then without the kite, and list in order.
You buy the kite and the bubbles. How much change?	$10 − $8 = $2.

20–32 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"Ways to pay $6 with $5 notes, $2 notes and $1 coins. I start with the biggest note: one $5. $1 is left, so one $1 coin. Now no $5: three $2, then two $2 and two $1, one $2 and four $1, no $2 and six $1. The $2 column went 3, 2, 1, 0, so I know I am finished: 5 ways."*

"Mei spent 80¢ on a drink, then 65¢ on a bun, and has 55¢ left. I work backwards from the end. Before the bun she had 55 + 65 = 120¢. Before the drink she had 120 + 80 = 200¢. That is $2."

"Change from $1 for 45¢: I count on like a shopkeeper. 45¢ and 5¢ is 50¢. 50¢ and 50¢ is $1. So the change is 5¢ + 50¢ = 55¢."

Worked example on the slides	Answer
Make a list in order: pay $6 with $5, $2 and $1. Start with the biggest note: one $5, then $1. No $5: three $2, two $2, one $2, no $2. Fill up with $1 coins.	5 ways.
Work backwards with money. Mei bought a drink for 80¢. Then she bought a bun for 65¢. She has 55¢ left. Before the bun: 55¢ + 65¢ = 120¢. Before the drink: 120¢ + 80¢ = ?	200¢ = $2.

SAY *"The key idea today is: in order, and backwards. To find every way, make a list in order: the most of the biggest note first, then one fewer each time. To find what someone had at first, work backwards and put back what was spent. Say it with me."*

DO Show the key idea slide at about minutes 31 to 32. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Mixing up dollars and cents: writing 345¢ as $34.5, or reading $3.05 as $3.50.
- Using a $20 note (there is no $20 note), or not starting with the biggest note that fits when finding the fewest.
- Listing ways with no order, so some ways are missed or the same way is written twice.

32–36 min Brain break

DO Shopkeeper stretch. The tutor calls an amount: reach up high for dollars, touch your toes for cents. "$3.50!" "75¢!" "205¢!" (that is $2.05: reach up!).

Water. Toilet break for anyone who needs it. Keep it to four minutes.

DO Story break (optional, 2 minutes): do the movement game for 2 minutes (32 to 34), then show the "Story break: Luca Pacioli" slide (34 to 36). Read the story aloud (about 30 seconds): "Luca Pacioli was a maths teacher in Italy about 500 years ago. In 1494 he printed a big maths book in Venice. He wrote it in Italian so more people could read it. It had a table of the coins, weights and measures used in different places. It also had a part on how a shop keeps its money records. He did not invent that method. He wrote down how Venice merchants already did it."

DO Then ask: "Would anyone like to try this one?" Try it: Keep a record like a merchant. You have $20. You spend $6 on a book and $5 on a cap. How much did you spend altogether, and how much is left? Volunteers only, never name a child, nobody is marked, passing is fine. SKIP the story break, without comment, if you are behind time or the group needs the full four minutes.

36–72 min Worksheet (pairs, then alone, then stretch)

DO Minute 36: give out the workbook. Read the title of Lesson 6 aloud. Say "Pairs first. Talk before you write."

Minutes 36 to 50: pairs work together on the first page (online, visit every shared breakout room by minute 47 so no pair works unseen for long; in Lessons 5, 6 and 10 a short teach slide comes first, 36 to 38). Minutes 50 to 52: check page 1 together. Minutes 52 to 57: short teaching for page 2.

Minutes 57 to 69: each child works alone on the second page. Minutes 69 to 72: check page 2 together. Think harder, Go further and the Investigation are open to any child at any time after question 3.

SAY *"Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking."*

DO Circulate and ask:

- Ask: "Is this in cents or in dollars? How do you know?"
- Ask: "What is the biggest note that is not too much?"
- Ask: "How do you know you have found every way?"

IF STUCK Hint 1: "Lay out real play money and count it."

Hint 2: "Make a table with a column for each note or coin. Start with as many of the biggest as you can."

Hint 3: "Start from the money that is left and put back what was spent."

DO A choice, not a track: after question 3 any child may carry on or try Think harder (the neutral line is printed on page 1). Invite the WHOLE group at once, never one child: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Never announce who is on which page, never name a child in front of the group, and do not split a pair unless both children agree. Children may move either way at any time. The Investigation is printed on the last core page, so every child sees it.

Think harder and Go further questions

Question	Answer
Workbook Think harder 1: Ben spent half of his money on a kite. Then he spent $4 on a ball. He had $6 left. How much money did Ben have at first? Tip: draw his money as a bar cut in half.	$20 ($6 + $4 = $10 is the half he kept after the kite, so the other half is also $10: $10 + $10 = $20).
Workbook Think harder 2: How many ways can you pay exactly $10 using $5 notes, $2 notes and $1 coins?	10 ways (1 + 3 + 6)
Workbook Think harder 3: Ali has 4 coins. They are worth $1.25 altogether. Find all the sets of 4 coins he could have.	2 sets: $1 + 10¢ + 10¢ + 5¢, and 50¢ + 50¢ + 20¢ + 5¢
Workbook Think harder 4: How many different ways can you pay exactly 40¢ using 5¢, 10¢ and 20¢ coins? Make a list in order, starting with the most 20¢ coins.	9 ways: two 20¢ (1 way); one 20¢ and 20¢ more (3 ways); no 20¢ (5 ways).

Workbook Go further 5: Siti had some money. She spent half of it on a book. Then she spent half of what was left on a pen. She had $5 left. How much money did she have at first?	$20 ($5 + $5 = $10 before the pen; $10 + $10 = $20)
Workbook Go further 6: Sweets cost 5¢ each and lollipops cost 20¢ each. Ben buys 8 things altogether and spends exactly $1. How many sweets and how many lollipops does he buy?	4 sweets and 4 lollipops (20¢ + 80¢ = $1)
Workbook Go further 7: Mei has one $10 note, one $5 note, one $2 note and one $1 coin. How many different amounts can she pay exactly, using one or more of them?	15 amounts: $1, $2, $3, $5, $6, $7, $8, $10, $11, $12, $13, $15, $16, $17, $18

These are the questions on the two Think harder pages in the workbook (Think harder, then Go further). More are in the separate Question Bank.

Investigation (several answers are right)	Model answer and what to look for
Workbook Investigation (last core page): Most coins. Use only 5¢, 10¢, 20¢ and 50¢ coins. (a) Pay 35¢ using the fewest coins. How many coins is that? (b) Try the other amounts from 5¢ to 95¢ (in steps of 5¢) and make a table of how many coins each needs. (c) Which amount needs the MOST coins, even when you use the fewest possible? Easy first step: pay 35¢ with the fewest coins from the 5¢, 10¢, 20¢ and 50¢ coins. Needs: play coins (5¢, 10¢, 20¢, 50¢), or paper and a pencil.	Fewest coins (5¢ to 95¢): 5:1, 10:1, 15:2, 20:1, 25:2, 30:2, 35:3, 40:2, 45:3, 50:1, 55:2, 60:2, 65:3, 70:2, 75:3, 80:3, 85:4, 90:3, 95:4 (brute-forced). (a) 35¢ = 20¢ + 10¢ + 5¢, 3 coins. (c) 85¢ and 95¢ need 4 coins each: 85¢ = 50¢ + 20¢ + 10¢ + 5¢ and 95¢ = 50¢ + 20¢ + 20¢ + 5¢. An amount ending in 5 needs a 5¢ coin, and the rest then needs 3 more coins. Look for: a systematic table, trying biggest coin first, and checking that no smaller number of coins works.

72–77 min Talk time

DO Bring the group back to a circle (or camera on, mics on). You choose volunteers first: "Who would like to share?" If nobody offers, ask the pairs to share what their partner said. Never cold-call a child by name. Any child may say "pass" at any time and that is always fine. Do not keep public tallies, scores or rankings (no counting who got most right, no "first to finish"). Invite the whole group to think about whether they would like a turn, never one child.

SAY *"Mei says: To pay $40 with the fewest notes, I use two $20 notes. What would you tell her?"*

"How did you make sure you found every way to pay?"

Discussion question	Listen for
Mei says: "To pay $40 with the fewest notes, I use two $20 notes." What would you tell her?	There is no $20 note in Singapore. $50 is too much, so we use $10 notes: $10 + $10 + $10 + $10, which is 4 notes. To find every way, start with the most of the biggest note and use one fewer each time, in a table.

77–80 min Investigation share (3 minutes)

DO Show the Investigation slide. The Investigation is on the last core page of the workbook and is open to every child from question 3 on. Ask the whole group: “Who would like to share one way they tried it? Passing is fine.” Volunteers only, no names, no public tallies. Several answers are right: praise the method, not the speed. If nobody has started, give pairs one minute for the easy first step and set the rest as take-home.

Investigation	Model answer and what to look for
Investigation: Most coins	Fewest coins (5¢ to 95¢): 5:1, 10:1, 15:2, 20:1, 25:2, 30:2, 35:3, 40:2, 45:3, 50:1, 55:2, 60:2, 65:3, 70:2, 75:3, 80:3, 85:4, 90:3, 95:4 (brute-forced). (a) 35¢ = 20¢ + 10¢ + 5¢, 3 coins. (c) 85¢ and 95¢ need 4 coins each: 85¢ = 50¢ + 20¢ + 10¢ + 5¢ and 95¢ = 50¢ + 20¢ + 20¢ + 5¢. An amount ending in 5 needs a 5¢ coin, and the rest then needs 3 more coins. Look for: a systematic table, trying biggest coin first, and checking that no smaller number of coins works.

80–87 min Exit ticket

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide.

Exit ticket	Answer
A pen costs $3.65. Pay exactly with the fewest notes and coins.	$2 + $1 + 50¢ + 10¢ + 5¢ (5 pieces)

87–89 min Looking back (2 minutes)

SAY *“Looking back takes 2 minutes. Find the small box called Looking back on your page. Finish one line, two lines or all three, and tick what helped you. You may pass on any line. This box is for you. If you tick "I am still stuck", your tutor will read it at the next lesson. Nothing is shared with the group.”*

DO Nothing is collected in front of the group and nobody has to read theirs out. The Looking back box is for the child. A paper box cannot be read live, so the tutor reads the journals at the next lesson, while stamping, and never by private message or photo. If a child ticked still stuck, start the next open session with a whole-group invitation (for example: "Anyone who would like another look at last lesson's idea, stay with me and we will do it together"), with a parent nearby; never single out one child. A child may volunteer one sentence, never by name unless they choose to. No public tallies. Praise the strategy, not the speed.

89–90 min Closing and stamps

SAY *“Great shopping, inquirers! You paid exactly, gave change and made lists in order. A good shopkeeper always checks, and so did you.”*

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide (this is for you only, never shared with the children). If a child marked D or H, plan two minutes of catch-up at the start of the next lesson.

Think harder and Go further

Find every way to pay exactly $10 using $5 notes, $2 notes and $1 coins. (There are 10.)

Answer key (workbook)

Q1 Curry puff $1.65: $1 + 50¢ + 10¢ + 5¢ (4). Notebook $4.30: $2 + $2 + 20¢ + 10¢ (4). Umbrella $18.75: $10 + $5 + $2 + $1 + 50¢ + 20¢ + 5¢ (7). Bicycle $76: $50 + $10 + $10 + $5 + $1 (5).

Q2 7 ways ($5, $2, $1): 1,1,1 / 1,0,3 / 0,4,0 / 0,3,2 / 0,2,4 / 0,1,6 / 0,0,8. Row H is spare: tell children "you may not need every row". (a) 7 ways (b) $5 + $1 + $1 + $1, and $2 + $2 + $2 + $2

Q3 (a) 35¢ (b) $24 change: $10 + $10 + $2 + $2 (there is no $20 note)

Q4 (a) No. The bag and the cap cost $14 + $8 = $22. She has $20, so she is $2 short. (b) She spends $17: book and cap ($9 + $8), or bag and pen ($14 + $3). Those are the only two pairs.

Q5 (a) 25¢ + 30¢ + 45¢ = 100¢ = $1.00 (Think harder 1 is the half-money puzzle with the kite and the ball: $20.)

Q6 3 ways: 4 erasers; 2 erasers and 5 stickers; 10 stickers. (1 or 3 erasers leave 150¢ or 50¢, which 20¢ stickers cannot make exactly.)

Story try-it (Luca Pacioli). Spent: $6 + $5 = $11. Left: $20 − $11 = $9. Check: $11 + $9 = $20.

Investigation (Most coins). Fewest coins (5¢ to 95¢): 5:1, 10:1, 15:2, 20:1, 25:2, 30:2, 35:3, 40:2, 45:3, 50:1, 55:2, 60:2, 65:3, 70:2, 75:3, 80:3, 85:4, 90:3, 95:4 (brute-forced). (a) 35¢ = 20¢ + 10¢ + 5¢, 3 coins. (c) 85¢ and 95¢ need 4 coins each: 85¢ = 50¢ + 20¢ + 10¢ + 5¢ and 95¢ = 50¢ + 20¢ + 20¢ + 5¢. An amount ending in 5 needs a 5¢ coin, and the rest then needs 3 more coins. Look for: a systematic table, trying biggest coin first, and checking that no smaller number of coins works.

Lesson 7: Fair Shares

Week 4 (first lesson) • Fractions • Workbook: Lesson 7

Objective	Recognise equal parts and name fractions with bottom numbers up to 12; compare unit fractions and like fractions; add and take away like fractions within one whole; solve fair-sharing stories with a bar.
Thinking skills	Checking for equal parts, comparing with the same whole, drawing bars, explaining
Materials	Paper strips of the same length (4 per child), paper squares, scissors, crayons, mini whiteboards.
Online set-up	Each child folds paper strips on camera; the tutor shows fraction bars on the slides and children colour their own bars.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0–1 min Opening

SAY *"Good morning, inquirers! Today we are cake cutters and pizza sharers. There is one rule: every share must be FAIR."*

"By the end you will name fair shares with fractions, tell which share is bigger, and explain how you know."

DO Greet each child by name. Show the lesson title slide. Read "Today we will..." aloud.

1–4 min Sneak-peek puzzles (3 minutes)

SAY *"These are sneak-peek puzzles. They are ideas we have not learnt yet, so guessing is welcome and so is saying "I do not know". Nobody is marked and nothing is read out. Start with question 1: use objects, a drawing or counting. Then try question 2. Show your answer on your own paper to your tutor on camera, or keep it to share if you want. If you would like help, the tutor will invite the whole group to ask in the open session, and a parent can stay nearby."*

DO Show the Sneak-peek puzzles slide at minute 1 (3 minutes: about 1 minute for question 1, then 2 minutes for question 2). Say out loud: these are ideas we have not learnt yet; guessing and "I do not know" are welcome; nobody is marked.

Answers stay private. Each child holds their own paper to the camera for you only (look, do not read out), or writes it on their own paper and shows only you on camera. Never raise whiteboards together and never show a board in front of the group. Never ask a named child "how do you know?" Ask for volunteers: "Would anyone like to tell us how they started?"

Question 1 is an access question: every child can try it with objects, a drawing or counting, so everyone has a first win. Question 2 is the thinking question. Do not mark either one. Click to show the answers at the end as "one way to think about it", without saying who had what.

Use what you see only to plan. Invite the WHOLE group at once, never one child: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Never announce who is on which page and never name a child in front of the group. Children choose, and may move either way at any time. Pairs are not split unless both children agree. Everybody joins the explore, the mini-lesson, the check-together slides and talk time.

Sneak-peek puzzle	Answer and what to listen for
Question 1 (access: objects, drawing or counting): Fold a paper strip (or draw a bar) into 2 equal parts. Colour 1 part. What fraction is coloured?	1/2 (one half). Listen for: Equal parts matter. Folding on camera is welcome. This is a first win for everyone.
Question 2 (thinking): Ali has 1/5 of a roti prata. Mei has 1/8 of the same prata. Who has more? How do you know?	Ali (1/5). The same prata cut into fewer parts gives bigger parts. Listen for: Many children say 1/8 because 8 is bigger. Draw two strips, one cut in 5 and one cut in 8, and compare. A drawing is a good answer.

DO The neutral line “After question 3 you may choose: carry on, or try Think harder.” is printed on page 1 of the workbook, next to the lesson title. Nobody is labelled and nothing is announced.

4–8 min Warm-up

DO Fair or not fair? Six snacks are cut into parts: thumbs up if all the parts are the same size, thumbs down if not.

SAY *“Look at each snack. Ali and Mei want to share it. Is it fair? Thumbs up for fair, thumbs down for not fair.”*

“Be ready to tell me why.”

DO Keep the game to two minutes (4 to 6). Then the quick questions (6 to 8): ask questions 1, 3 and 5; questions 2 and 4 are spare. Children answer by thumbs, a whisper to a partner. Do not name a child.

Ask	Expect
A pizza is cut into 4 equal slices. What is 1 slice called?	One quarter, 1/4
How many halves make 1 whole?	2
How many thirds make 1 whole?	3
A cake is cut into 5 equal pieces. What is 1 piece called?	One fifth, 1/5
2 of 7 equal parts are shaded. What fraction is shaded?	2/7

8–20 min Explore (in pairs, with objects)

SAY *"Before we fold: which will be bigger, one quarter of a strip or one eighth of the same strip? Show me 4 fingers for a quarter or 8 fingers for an eighth."*

DO 1. Take 4 paper strips that are the same length.

2. Keep one whole. Fold one in half, one into quarters and one into eighths.

3. Open them out. Write the fraction on every part.

4. Line them up. Then try to fold a strip into 3 equal parts.

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK Stuck folding: match the two ends exactly, then press the fold flat.

Not sure what to write: "How many equal parts are there? That number goes at the bottom."

Finished early: can you fold a strip into 3 equal parts? How do you check they are equal?

DO Go further (a second what-if, so nobody waits): show the Explore: go further slide or post it in the chat when pairs go to their rooms. Any child may try it any time, as a choice. Never name who does. Question: fold a strip into 6 equal parts. How many sixths make a half? How many sixths make a third? Answer: 3 sixths make a half and 2 sixths make a third.

DO At about minute 14, say "Pencils down" and bring the group back together. Take the three questions below, one at a time (about 6 minutes).

Ask	Expect
Which part is bigger, 1/4 or 1/8? How do you know?	1/4: the same strip cut into fewer parts, so each part is longer.
How many eighths fit on one quarter? How many quarters make 1 whole?	2 eighths; 4 quarters.
How did you fold 3 equal parts? How do you know they are equal?	Fold and adjust until the parts match; check by laying one part on another.

20–32 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"Before I name a fraction I check the parts are equal. This bar has 5 parts, all the same size, so the parts are fifths: 5 goes at the bottom. 3 parts are shaded: 3 goes on top. Three fifths."*

"One half or one sixth: which is bigger? Same whole. One half is 1 of 2 parts, one sixth is 1 of 6 parts. More parts means smaller parts, so one half is bigger."

"Two ninths and four ninths: ninths are the name of the parts, like apples. 2 ninths and 4 ninths are 6 ninths. The parts do not change size, so the bottom number stays 9."

Worked example on the slides	Answer

Name the fraction. A bar has 5 equal parts and 3 are shaded. Check the parts are equal, count all the parts (bottom number), then count the shaded parts (top number). What fraction is shaded? What fraction is not?	3/5 shaded, 2/5 not shaded; together 5/5, 1 whole.
Compare with bars. Bars of the same length cut into halves, thirds and sixths, one part shaded in each. Which is bigger: 1/2, 1/3 or 1/6? Then: which is bigger, 2/7 or 5/7?	1/6 < 1/3 < 1/2 (more equal parts, smaller parts). 5/7 > 2/7 (same-size parts, more of them).

SAY *"The key idea today is: equal parts. A fraction names equal parts of a whole: the bottom number is how many equal parts, the top number is how many we take. Say it with me."*

DO Show the key idea slide at about minutes 31 to 32. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Naming a fraction when the parts are not equal: circling B or D in Q1.
- Thinking 1/6 is bigger than 1/4 because 6 is bigger than 4 (Q3).
- Adding the bottom numbers too: 2/5 + 1/5 written as 3/10 instead of 3/5 (Q4).

32–36 min Brain break

DO Fraction turns. Stand up and turn a whole turn, a half turn, a quarter turn, three quarter turns, then freeze.

Water. Toilet break for anyone who needs it. Keep it to four minutes.

DO Story break (optional, 2 minutes): do the movement game for 2 minutes (32 to 34), then show the "Story break: Ahmes the scribe" slide (34 to 36). Read the story aloud (about 30 seconds): "Ahmes was a scribe in ancient Egypt, more than 3,500 years ago. He wrote out a long maths roll on papyrus. It has 84 problems and tables for dividing numbers and handling fractions. Egyptian maths often wrote fractions with a 1 on top, like 1/3 and 1/4. Ahmes did not say the ideas were his own. He said he was copying an older work. The roll is now kept in the British Museum in London, and is called the Rhind Papyrus."

DO Then ask: "Would anyone like to try this one?" Try it: Share 1 loaf of bread equally among 4 children. What fraction of the loaf does each child get? What fraction do 3 of the children get together? Volunteers only, never name a child, nobody is marked, passing is fine. SKIP the story break, without comment, if you are behind time or the group needs the full four minutes.

36–72 min Worksheet (pairs, then alone, then stretch)

DO Minute 36: give out the workbook. Read the title of Lesson 7 aloud. Say "Pairs first. Talk before you write."

Minutes 36 to 50: pairs work together on the first page (online, visit every shared breakout room by minute 47 so no pair works unseen for long; in Lessons 5, 6 and 10 a short teach slide comes first, 36 to 38). Minutes 50 to 52: check page 1 together. Minutes 52 to 57: short teaching for page 2. Minutes 57 to 69: each child works alone on the second page. Minutes 69 to 72: check page 2

together. Think harder, Go further and the Investigation are open to any child at any time after question 3.

SAY *"Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking."*

DO Circulate and ask:

- Ask: "Are all the parts the same size? How do you know?"
- Ask: "What does the bottom number tell us? What does the top number tell us?"
- Ask: "Did you draw both wholes the same length?"

IF STUCK Hint 1: "Count all the equal parts first. That number goes at the bottom."

Hint 2: "Fold or draw a strip to check which part is bigger."

Hint 3: "Colour the parts on a bar, then count them."

DO A choice, not a track: after question 3 any child may carry on or try Think harder (the neutral line is printed on page 1). Invite the WHOLE group at once, never one child: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Never announce who is on which page, never name a child in front of the group, and do not split a pair unless both children agree. Children may move either way at any time. The Investigation is printed on the last core page, so every child sees it.

Think harder and Go further questions

Question	Answer
Workbook Think harder 1: Write these in order, smallest first: 1/3, 1/12, 1/2, 1/5.	1/12, 1/5, 1/3, 1/2
Workbook Think harder 2: A ribbon is cut into 9 equal pieces. Mei uses 4 pieces and Siti uses 3 pieces. What fraction of the ribbon is left?	2/9 (4 + 3 = 7 pieces used, 9 − 7 = 2 left)
Workbook Think harder 3: 2/7 + ? = 1 whole	5/7
Workbook Think harder 4: Ali says: "Half of my pizza is bigger than half of your pizza." Can he be right? Explain.	Yes, if his pizza is bigger. Halves of different-size wholes are different sizes.
Workbook Go further 5: A cake is cut into 12 equal pieces. Mei gets 4/12. Ali and Ben share the rest, but Ali gets 2/12 more than Ben. What fraction does Ben get?	3/12 (and Ali gets 5/12): 8 pieces are shared; take away Ali's extra 2, share 6 as 3 and 3.
Workbook Go further 6: A paper strip is folded in half, then in half again, then in half again. When you open it, how many equal parts are there? What fraction is each part?	8 equal parts; each is 1/8 (1 fold: 2 parts, 2 folds: 4, 3 folds: 8).

Workbook Go further 7: Siti eats 1/2 of a pizza. Then Ravi eats 1/2 of what is left. What fraction of the whole pizza is left now?	1/4. After Siti, half is left; Ravi eats half of that half, which is a quarter of the pizza, so a quarter is left.

These are the questions on the two Think harder pages in the workbook (Think harder, then Go further). More are in the separate Question Bank.

Investigation (several answers are right)	Model answer and what to look for
Workbook Investigation (last core page): Fraction strips. (a) Cover a HALF strip exactly with pieces of ONE kind only. Which kinds work? How many pieces of each? (b) Do the same for a THIRD strip and a QUARTER strip. (c) Can thirds ever cover a half exactly? Why not? Easy first step: lay the two quarter strips on top of the half strip. Do they fit exactly? Needs: the fraction strips from the cut-out page (halves, thirds, quarters, sixths, eighths).	(a) Half: 2 quarters, 3 sixths or 4 eighths (not thirds). (b) Third: 2 sixths only. Quarter: 2 eighths only (checked with exact fractions). (c) No: one third is shorter than a half but two thirds is longer, so no number of thirds fits exactly. Look for: children lay the strips side by side and say “fits exactly”. Keep to the strips and drawings: do not teach “equivalent fractions” as a rule (that is P3).

72–77 min Talk time

DO Bring the group back to a circle (or camera on, mics on). You choose volunteers first: "Who would like to share?" If nobody offers, ask the pairs to share what their partner said. Never cold-call a child by name. Any child may say "pass" at any time and that is always fine. Do not keep public tallies, scores or rankings (no counting who got most right, no "first to finish"). Invite the whole group to think about whether they would like a turn, never one child.

SAY *“Is 1/2 always bigger than 1/4? Can you think of a time when it is not?”*

“Why does the bottom number stay the same when we add 2/9 and 4/9?”

Discussion question	Listen for
Is 1/2 always bigger than 1/4? When could it not be?	Only when the wholes are the same size: half a small cupcake is less than a quarter of a big cake. When we add ninths, the parts are still ninths, so only the number of parts changes.

77–80 min Investigation share (3 minutes)

DO Show the Investigation slide. The Investigation is on the last core page of the workbook and is open to every child from question 3 on. Ask the whole group: “Who would like to share one way they tried it? Passing is fine.” Volunteers only, no names, no public tallies. Several answers are right: praise the method, not the speed. If nobody has started, give pairs one minute for the easy first step and set the rest as take-home.

Investigation	Model answer and what to look for

Investigation: Fraction strips	(a) Half: 2 quarters, 3 sixths or 4 eighths (not thirds). (b) Third: 2 sixths only. Quarter: 2 eighths only (checked with exact fractions). (c) No: one third is shorter than a half but two thirds is longer, so no number of thirds fits exactly. Look for: children lay the strips side by side and say “fits exactly”. Keep to the strips and drawings: do not teach “equivalent fractions” as a rule (that is P3).

80–87 min Exit ticket

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide.

Exit ticket	Answer
Which is bigger, 1/7 or 1/9? Then work out 2/9 + 5/9.	1/7 (the same whole cut into fewer parts gives bigger parts). 2/9 + 5/9 = 7/9.

87–89 min Looking back (2 minutes)

SAY *“Looking back takes 2 minutes. Find the small box called Looking back on your page. Finish one line, two lines or all three, and tick what helped you. You may pass on any line. This box is for you. If you tick "I am still stuck", your tutor will read it at the next lesson. Nothing is shared with the group.”*

DO Nothing is collected in front of the group and nobody has to read theirs out. The Looking back box is for the child. A paper box cannot be read live, so the tutor reads the journals at the next lesson, while stamping, and never by private message or photo. If a child ticked still stuck, start the next open session with a whole-group invitation (for example: "Anyone who would like another look at last lesson's idea, stay with me and we will do it together"), with a parent nearby; never single out one child. A child may volunteer one sentence, never by name unless they choose to. No public tallies. Praise the strategy, not the speed.

89–90 min Closing and stamps

SAY *“You checked for equal parts, compared fractions with bars and explained why. That is fair and careful maths.”*

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide (this is for you only, never shared with the children). If a child marked D or H, plan two minutes of catch-up at the start of the next lesson.

Think harder and Go further

Find all the ways to add two fractions with bottom number 8 to make 1 whole (1/8 + 7/8, 2/8 + 6/8, 3/8 + 5/8, 4/8 + 4/8: 4 ways).

Answer key (workbook)

Q1 A, C, E and F are cut into quarters. (a) B or D: their 4 parts are not the same size, so a part is not a quarter. F is the hardest: all four small triangles are the same size, even though the middle one is upside

down.

Q2 (a) shaded 3/8, not shaded 5/8 (b) 5/6 and 1/6 (c) 7/10 and 3/10

Q3 (a) 1/3; 7/9 (b) 1/10, 1/5, 1/4, 1/2 (c) No. Cutting the same whole into 6 equal parts makes smaller parts than cutting it into 4, so 1/6 is smaller than 1/4.

Q4 (a) 3/5 (b) 7/8 (c) 9/12 (d) 5/6 (e) 4/10

Q5 (a) 5/8 (b) 3/8 (c) Mei, by 1/8

Q6 Siti. The pizzas are the same size and one quarter covers 2 eighths on the bars; 3 eighths is more than 2 eighths. (Equivalent fractions are a P3 topic: show this with bars only and do not write 1/4 = 2/8 as a rule.)

Story try-it (Ahmes the scribe). Each child gets 1/4 of the loaf. Three children get 3/4 together.

Investigation (Fraction strips). (a) Half: 2 quarters, 3 sixths or 4 eighths (not thirds). (b) Third: 2 sixths only. Quarter: 2 eighths only (checked with exact fractions). (c) No: one third is shorter than a half but two thirds is longer, so no number of thirds fits exactly. Look for: children lay the strips side by side and say “fits exactly”. Keep to the strips and drawings: do not teach “equivalent fractions” as a rule (that is P3).

Lesson 8: Shape Builders

Week 4 (second lesson) • 2D and 3D Shapes • Workbook: Lesson 8

Objective	Name semicircles and quarter circles and use them to make and break up shapes; count every square and triangle in a figure by working in order; count the cubes in a stack, including hidden ones, layer by layer; decide which nets fold into a cube.
Thinking skills	Spatial reasoning, systematic counting, folding in your head, explaining
Materials	Paper circles (2 per child), scissors, about 12 linking cubes per child (or sugar cubes), the cube-net cut-out page (nets A to F, near the back of the workbook), mini whiteboards.
Online set-up	Children cut circles and build stacks on camera; the six nets A to F are on a cut-out page near the back of the workbook (print and cut them before the lesson), or cut a net from an empty box. No linking cubes at home? Use sugar cubes, dice or small boxes, or draw each stack layer by layer.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0–1 min Opening

SAY *"Good morning, inquirers! Today we are shape builders. We will cut circles into pieces, hunt for hidden squares and triangles, count cubes we cannot even see, and fold flat paper into a cube."*

"The secret for every puzzle today: count in order, so nothing is missed and nothing is counted twice."

DO Greet each child by name. Show the lesson title slide. Read "Today we will..." aloud.

1–4 min Sneak-peek puzzles (3 minutes)

SAY *"These are sneak-peek puzzles. They are ideas we have not learnt yet, so guessing is welcome and so is saying "I do not know". Nobody is marked and nothing is read out. Start with question 1: use objects, a drawing or counting. Then try question 2. Show your answer on your own paper to your tutor on camera, or keep it to share if you want. If you would like help, the tutor will invite the whole group to ask in the open session, and a parent can stay nearby."*

DO Show the Sneak-peek puzzles slide at minute 1 (3 minutes: about 1 minute for question 1, then 2 minutes for question 2). Say out loud: these are ideas we have not learnt yet; guessing and "I do not know" are welcome; nobody is marked.

Answers stay private. Each child holds their own paper to the camera for you only (look, do not read out), or writes it on their own paper and shows only you on camera. Never raise whiteboards together and never show a board in front of the group. Never ask a named child "how do you know?" Ask for volunteers: "Would anyone like to tell us how they started?"

Question 1 is an access question: every child can try it with objects, a drawing or counting, so everyone has a first win. Question 2 is the thinking question. Do not mark either one. Click to show

the answers at the end as "one way to think about it", without saying who had what.

Use what you see only to plan. Invite the WHOLE group at once, never one child: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Never announce who is on which page and never name a child in front of the group. Children choose, and may move either way at any time. Pairs are not split unless both children agree. Everybody joins the explore, the mini-lesson, the check-together slides and talk time.

Sneak-peek puzzle	Answer and what to listen for
Question 1 (access: objects, drawing or counting): A stack has 2 layers. Each layer has 2 rows of 3 cubes. How many cubes are in the stack? (Build it with cubes or draw it.)	12 cubes (6 in each layer, 2 layers). Listen for: Counting one layer first, then doubling, is a good method. This is a first win for everyone.
Question 2 (thinking): A shape is made of 2 semicircles and 1 quarter circle. If you cut every piece into quarter circles, how many quarter circles do you get?	5 (each semicircle is 2 quarters: 2 + 2 + 1). Listen for: Listen for "a semicircle is half a circle, which is 2 quarters". Drawing the pieces and cutting them is fine.

DO The neutral line “After question 3 you may choose: carry on, or try Think harder.” is printed on page 1 of the workbook, next to the lesson title. Nobody is labelled and nothing is announced.

4–8 min Warm-up

DO Name the solid: a dice, a tissue box, a tin can, a party hat, a drum and a traffic cone. Make the shape with your hands, then name it.

SAY *“I will show you something from home. Make its shape with your hands: a cube, a cuboid, a cylinder or a cone. Then we say its name together.”*

DO Keep the game to two minutes (4 to 6). Then the quick questions (6 to 8): ask questions 1, 3 and 5; questions 2 and 4 are spare. Children answer by thumbs, a whisper to a partner. Do not name a child.

Ask	Expect
How many faces does a cube have?	6
How many flat faces does a cylinder have?	2
Cut a circle into 2 equal parts. What is each part called?	A semicircle
How many quarter circles make 1 whole circle?	4
Which solid has a tip and can roll?	A cone

8–20 min Explore (in pairs, with objects)

SAY *"Before we cut: if I cut a circle in half, and then cut one half in half again, how many pieces will I have? What will they look like?"*

DO 1. Take 2 paper circles that are the same size.

2. Fold one in half and cut along the fold.

3. Fold the other in half, then in half again. Open it and cut along both folds.

4. Use your pieces to make new shapes. Name them.

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK Stuck folding: fold the circle so the edges match exactly, then press the fold flat.

Not sure of a name: "Count the straight sides. One straight side: semicircle. Two straight sides: quarter circle."

Finished early: find 3 different ways to make a whole circle from your pieces.

DO Go further (a second what-if, so nobody waits): show the Explore: go further slide or post it in the chat when pairs go to their rooms. Any child may try it any time, as a choice. Never name who does. Question: you have 2 semicircles and 3 quarter circles. How many whole circles can you make? What is left over? Answer: 7 quarter circles in all: 1 whole circle (4 quarters) and 3 quarters left over.

DO At about minute 14, say "Pencils down" and bring the group back together. Take the three questions below, one at a time (about 6 minutes).

Ask	Expect
How many quarter circles cover a semicircle exactly?	2
Find 3 different ways to make a whole circle from your pieces.	2 semicircles; 4 quarter circles; 1 semicircle and 2 quarter circles.
How many straight sides has a quarter circle? Are they the same length?	2, and yes: they meet at a square corner.

20–32 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"To name a piece I count its straight sides. One straight side and one curved side: a semicircle. Two straight sides that meet at a square corner, and one curved side: a quarter circle. Turning it round does not change its name."*

"Squares hide inside bigger squares. I count by size: 1 by 1 first, then I slide a 2 by 2 window and count the places it fits, then 3 by 3. Then I add them up."

"For cubes I count layer by layer. Cubes cannot float, so if I see a cube on top, there must be a cube under it, even if I cannot see it."

Worked example on the slides	Answer
Find the pieces. A mushroom is made of a semicircle cap, a square stem and 2 quarter circles of grass. Count one kind at a time: 1 straight and 1 curved side, then 2 straight sides and 1 curved side, then 4 straight sides. How many pieces of each kind?	Semicircle 1, quarter circles 2, square 1: 4 pieces.
Count every square. A grid 3 squares long and 3 squares high. Count by size: 1 by 1, then slide a 2 by 2 window, then the whole grid. How many squares altogether?	9 + 4 + 1 = 14 squares.

SAY *"The key idea today is: count in order. Sort what you count, by size, by the line it sits on or by layer, so you never miss one or count one twice. Say it with me."*

DO Show the key idea slide at about minutes 31 to 32. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Counting only the 1 by 1 squares in Q2a (16 instead of 30).
- Counting only the cubes you can see in Q3 and forgetting the hidden cubes underneath and behind.
- Thinking any 6 joined squares fold into a cube (Q4): B, D and F do not (cutting out and folding shows why).

32–36 min Brain break

DO Be a shape! Make a semicircle, a cone, a cube and a cylinder with your body when the tutor calls it.

Water. Toilet break for anyone who needs it. Keep it to four minutes.

DO Story break (optional, 2 minutes): do the movement game for 2 minutes (32 to 34), then show the "Story break: Alicia Boole Stott" slide (34 to 36). Read the story aloud (about 30 seconds): "Alicia Boole Stott was born in Cork, Ireland, in 1860. She never went to university. She learned from her mother and from geometry books. As a young woman she played with wooden cubes to think about shapes in new ways. Later she cut and joined cardboard to build models of very unusual shapes that are hard to draw. A Dutch mathematician, Pieter Schoute, saw her models and they worked together for many years."

DO Then ask: "Would anyone like to try this one?" Try it: A cube is built from flat square pieces. How many square faces does a cube have? How many edges does it have? Volunteers only, never name a child, nobody is marked, passing is fine. SKIP the story break, without comment, if you are behind time or the group needs the full four minutes.

36–72 min Worksheet (pairs, then alone, then stretch)

DO Minute 36: give out the workbook. Read the title of Lesson 8 aloud. Say "Pairs first. Talk before you write."

Minutes 36 to 50: pairs work together on the first page (online, visit every shared breakout room by minute 47 so no pair works unseen for long; in Lessons 5, 6 and 10 a short teach slide comes first, 36 to 38). Minutes 50 to 52: check page 1 together. Minutes 52 to 57: short teaching for page 2. Minutes 57 to 69: each child works alone on the second page. Minutes 69 to 72: check page 2 together. Think harder, Go further and the Investigation are open to any child at any time after question 3.

SAY *"Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking."*

DO Circulate and ask:

- Ask: "How many straight sides does it have? So what is it called?"
- Ask: "Where else can your window fit? Have you tried every place?"
- Ask: "What is holding up that top cube?"

IF STUCK Hint 1: "Count one kind or one size at a time, and tick each one."

Hint 2: "Build it with cubes, or cut out the net and fold it."

Hint 3: "For a net, pick a bottom square and fold the others up one at a time."

DO A choice, not a track: after question 3 any child may carry on or try Think harder (the neutral line is printed on page 1). Invite the WHOLE group at once, never one child: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Never announce who is on which page, never name a child in front of the group, and do not split a pair unless both children agree. Children may move either way at any time. The Investigation is printed on the last core page, so every child sees it.

Think harder and Go further questions

Question	Answer
Workbook Think harder 1: A stack of cubes has a bottom layer 4 cubes long and 3 cubes deep, a middle layer 3 cubes long and 2 cubes deep, and 1 cube on top. How many cubes are in the stack? Draw the layers.	19 (12 + 6 + 1)
Workbook Think harder 2: How many squares are in a grid 3 squares long and 2 squares high?	8 (6 small + 2 made of 4 small ones)
Workbook Think harder 3: Cut out nets E and F from the cut-out page and try folding them. Which one folds up into a cube? How do you know?	E folds into a cube (it is a staircase of squares, one of the 11 cube nets). F does not: the row of 5 squares wraps round and the 1st and 5th squares land on the same face.
Workbook Think harder 4: Six squares are joined in a straight line. Can they fold into a cube? Explain.	No. When you fold the line round, the 5th square lands on the same face as the 1st and the 6th on the 2nd, and the two ends are left open.

Workbook Go further 5: A pyramid of cubes has a top layer of 1 cube, then a layer 2 by 2, then a layer 3 by 3, then a bottom layer 4 by 4. How many cubes are there?	30 (1 + 4 + 9 + 16)
Workbook Go further 6: How many squares can you find in a grid 5 squares long and 5 squares high?	55 (25 + 16 + 9 + 4 + 1)
Workbook Go further 7: On a dice, opposite faces add up to 7. You can see 3 faces showing 1, 2 and 3. What numbers are on the 3 faces you cannot see? What do they add up to?	6, 5 and 4; they add up to 15.

These are the questions on the two Think harder pages in the workbook (Think harder, then Go further). More are in the separate Question Bank.

Investigation (several answers are right)	Model answer and what to look for
Workbook Investigation (last core page): Squares joined edge to edge. Squares must touch along a whole side. A shape that is turned round or flipped over counts as the SAME shape. (a) Find ALL the different shapes you can make with 3 squares. (b) Now find ALL the different shapes you can make with 4 squares. Draw each one on the grid. (c) How do you know you have found them all? Easy first step: join 2 squares edge to edge. How many different shapes can you make? Then add a third square. Needs: square tiles or small squares of paper, and the grid on this page.	(a) 2 shapes: a straight line and an L. (b) 5 shapes: a straight line of 4, a 2 x 2 square, an L, a T and a zigzag (S shape). Brute-forced: 1, 1, 2, 5 shapes for 1, 2, 3, 4 squares (turns and flips count as the same). If a child counts flips as different, they find 7 (L and its mirror, S and its mirror): praise this and explain the rule. (c) Look for a system: start with the straight line, then move one square, and check each new shape against the others by turning the tile.

72–77 min Talk time

DO Bring the group back to a circle (or camera on, mics on). You choose volunteers first: "Who would like to share?" If nobody offers, ask the pairs to share what their partner said. Never cold-call a child by name. Any child may say "pass" at any time and that is always fine. Do not keep public tallies, scores or rankings (no counting who got most right, no "first to finish"). Invite the whole group to think about whether they would like a turn, never one child.

SAY *"How can you be sure you counted every cube in a stack, even the ones you cannot see?"*

"Why can 6 squares in a 2 by 3 rectangle not fold into a cube?"

Discussion question	Listen for

How can you be sure you counted every cube in a stack, even the ones you cannot see?	Count layer by layer; every cube on top sits on a cube below, so the hidden ones must be there. In a 2 by 3 rectangle, 4 squares meet at a corner, but only 3 faces of a cube meet at a corner, so two squares end up on the same face.

77–80 min Investigation share (3 minutes)

DO Show the Investigation slide. The Investigation is on the last core page of the workbook and is open to every child from question 3 on. Ask the whole group: “Who would like to share one way they tried it? Passing is fine.” Volunteers only, no names, no public tallies. Several answers are right: praise the method, not the speed. If nobody has started, give pairs one minute for the easy first step and set the rest as take-home.

Investigation	Model answer and what to look for
Investigation: Squares joined edge to edge	(a) 2 shapes: a straight line and an L. (b) 5 shapes: a straight line of 4, a 2 x 2 square, an L, a T and a zigzag (S shape). Brute-forced: 1, 1, 2, 5 shapes for 1, 2, 3, 4 squares (turns and flips count as the same). If a child counts flips as different, they find 7 (L and its mirror, S and its mirror): praise this and explain the rule. (c) Look for a system: start with the straight line, then move one square, and check each new shape against the others by turning the tile.

80–87 min Exit ticket

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide.

Exit ticket	Answer
How many cubes are in a stack with a bottom layer 3 cubes long and 2 cubes deep, and 1 cube on top?	7 (6 + 1)

87–89 min Looking back (2 minutes)

SAY *“Looking back takes 2 minutes. Find the small box called Looking back on your page. Finish one line, two lines or all three, and tick what helped you. You may pass on any line. This box is for you. If you tick "I am still stuck", your tutor will read it at the next lesson. Nothing is shared with the group.”*

DO Nothing is collected in front of the group and nobody has to read theirs out. The Looking back box is for the child. A paper box cannot be read live, so the tutor reads the journals at the next lesson, while stamping, and never by private message or photo. If a child ticked still stuck, start the next open session with a whole-group invitation (for example: "Anyone who would like another look at last lesson's idea, stay with me and we will do it together"), with a parent nearby; never

single out one child. A child may volunteer one sentence, never by name unless they choose to. No public tallies. Praise the strategy, not the speed.

89–90 min Closing and stamps

SAY *"You named pieces by their sides, counted in order and folded nets in your head. That is real shape building."*

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide (this is for you only, never shared with the children). If a child marked D or H, plan two minutes of catch-up at the start of the next lesson.

Think harder and Go further

How many squares are in a grid 5 squares long and 5 squares high? (25 + 16 + 9 + 4 + 1 = 55)

Answer key (workbook)

Q1 Square 2, triangle 3, semicircle 4, quarter circle 2. (a) semicircles (b) 11 pieces (c) 3 circles: 3 semicircles are 6 quarter circles, 6 + 6 = 12 quarter circles, and 4 quarter circles make a circle.

Q2 (a) 1 by 1: 16, 2 by 2: 9, 3 by 3: 4, 4 by 4: 1; 30 squares (b) 6 on the bottom line + 6 on the line across = 12 triangles

Q3 (a) A 5, B 8, C 14 (9 + 4 + 1) (b) 13 more cubes (27 − 14)

Q4 (a) A and C fold into a cube; B and D do not (children cut out and fold to check). Nets E and F are Think harder 3: E folds, F does not (b) the new square goes under any one of the 4 squares in the row (4 possible places)

Story try-it (Alicia Boole Stott). A cube has 6 square faces and 12 edges.

Investigation (Squares joined edge to edge). (a) 2 shapes: a straight line and an L. (b) 5 shapes: a straight line of 4, a 2 x 2 square, an L, a T and a zigzag (S shape). Brute-forced: 1, 1, 2, 5 shapes for 1, 2, 3, 4 squares (turns and flips count as the same). If a child counts flips as different, they find 7 (L and its mirror, S and its mirror): praise this and explain the rule. (c) Look for a system: start with the straight line, then move one square, and check each new shape against the others by turning the tile.

Lesson 9: Measure Masters

Week 5 (first lesson) • Length, Mass, Volume and Time • Workbook: Lesson 9

Objective	Solve balance-scale puzzles in g and kg by swapping things that balance; solve pouring puzzles with litre jugs by keeping a table of every pour; join and cut lengths in m and cm (overlaps, pieces and cuts); tell the time to 5 minutes with a.m. and p.m. and find start times, end times and how long, also across the hour.
Thinking skills	Reasoning with equal amounts (swapping), working step by step, keeping a table, counting on in time, checking
Materials	A coat hanger, two paper cups, string, a glue stick, a spoon and about 20 20¢ coins (a home-made balance; backup: a ruler balanced on a pencil); for the jug puzzles two cups or bottles of different size (about 3 and 5 units), a jug of dry rice or sand (or water), and tape marking 3 and 5 units; a cardboard clock with hands; mini whiteboards.
Online set-up	Children build the coat-hanger balance before the lesson and hold it up to the camera. No coat-hanger balance at home? The tutor balances a ruler on a pencil on camera and pairs share one view. The tutor shares the slide balances, jugs and clocks and annotates on screen. Each child needs a cardboard or paper-plate clock. For the pouring puzzles use dry rice or sand rather than water when online, and let children act out each pour with the two containers.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0–1 min Opening

SAY *"Good morning, Measure Masters! Today we measure how long, how heavy, how much water and how much time."*

"Some puzzles today have no ruler, no scale with numbers and no marks on the jugs. We will use our heads: notice it, test it, explain why."

DO Greet each child by name. Show the lesson title slide. Read "Today we will…" aloud.

1–4 min Sneak-peek puzzles (3 minutes)

SAY *"These are sneak-peek puzzles. They are ideas we have not learnt yet, so guessing is welcome and so is saying "I do not know". Nobody is marked and nothing is read out. Start with question 1: use objects, a drawing or counting. Then try question 2. Show your answer on your own paper to your tutor on camera, or keep it to share if you want. If you would like help, the tutor will invite the whole group to ask in the open session, and a parent can stay nearby."*

DO Show the Sneak-peek puzzles slide at minute 1 (3 minutes: about 1 minute for question 1, then 2 minutes for question 2). Say out loud: these are ideas we have not learnt yet; guessing and "I do not know" are welcome; nobody is marked.

Answers stay private. Each child holds their own paper to the camera for you only (look, do not read out), or writes it on their own paper and shows only you on camera. Never raise whiteboards together and never show a board in front of the group. Never ask a named child "how do you know?" Ask for volunteers: "Would anyone like to tell us how they started?"

Question 1 is an access question: every child can try it with objects, a drawing or counting, so everyone has a first win. Question 2 is the thinking question. Do not mark either one. Click to show the answers at the end as "one way to think about it", without saying who had what.

Use what you see only to plan. Invite the WHOLE group at once, never one child: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Never announce who is on which page and never name a child in front of the group. Children choose, and may move either way at any time. Pairs are not split unless both children agree. Everybody joins the explore, the mini-lesson, the check-together slides and talk time.

Sneak-peek puzzle	Answer and what to listen for
Question 1 (access: objects, drawing or counting): You have a 3 ℓ jug and a 5 ℓ jug. You fill the 5 ℓ jug and pour from it into the 3 ℓ jug until the 3 ℓ jug is full. How much water is left in the 5 ℓ jug? (Try it with two cups and rice, or draw it.)	2 ℓ (5 - 3 = 2). Listen for: One pour is enough. This is a first win for everyone and previews the pour table.
Question 2 (thinking): A melon balances 2 apples. One apple balances 3 plums. How many plums balance the melon? How do you know?	6 plums (3 + 3). Listen for: Listen for swapping: the melon can be swapped for 2 apples, and each apple for 3 plums. Drawing the scale is a good method.

DO The neutral line “After question 3 you may choose: carry on, or try Think harder.” is printed on page 1 of the workbook, next to the lesson title. Nobody is labelled and nothing is announced.

4–8 min Warm-up

DO Which unit? Show six things one at a time (a door, a pencil, a bag of rice, a strawberry, a bucket of water, brushing teeth). Children write m, cm, kg, g, ℓ or minutes on whiteboards and explain.

SAY *“I will show you something. Write the unit you would use to measure it on your whiteboard: m, cm, kg, g, ℓ or minutes. Show your tutor on camera or keep it to share if you want.”*

“Be ready to say why. Why not grams for a bag of rice?”

DO Keep the game to two minutes (4 to 6). Then the quick questions (6 to 8): ask questions 1, 3 and 5; questions 2 and 4 are spare. Children answer by thumbs, a whisper to a partner. Do not name a child.

Ask	Expect
35 cm + 40 cm = ?	75 cm
The minute hand points to 6. How many minutes past the hour?	30 minutes

A 5 ℓ bucket has 2 ℓ of water. How much more fills it?	3 ℓ
Lunch at 12:30. a.m. or p.m.?	p.m.
One bag of rice is 5 kg. How heavy are 2 bags?	10 kg

8–20 min Explore (in pairs, with objects)

SAY *"Look at your balance. A glue stick goes in one cup. How many 20¢ coins will balance it? Write your guess before you test."*

DO 1. Hang two paper cups from a coat hanger. Check it is level when the cups are empty.

2. Put a glue stick in one cup. Guess: how many 20¢ coins will balance it?

3. Add coins one at a time until the hanger is level. Then try a spoon.

4. Which is heavier, the glue stick or the spoon? How do you know?

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK Stuck: check the hanger is level with both cups empty before you start.

Coins keep sliding: put them in the cup one at a time and stop when the hanger is level.

Finished early: find something that balances exactly the same number of coins as your glue stick.

DO Go further (a second what-if, so nobody waits): show the Explore: go further slide or post it in the chat when pairs go to their rooms. Any child may try it any time, as a choice. Never name who does. Question: one glue stick balances 8 coins. A pencil case balances 3 glue sticks. How many coins balance the pencil case? If an eraser balances 4 coins, how many erasers balance the pencil case? Answer: 24 coins (3 x 8). 6 erasers (24 / 4).

DO At about minute 14, say "Pencils down" and bring the group back together. Take the three questions below, one at a time (about 6 minutes).

Ask	Expect
How many coins balanced your glue stick? Your spoon?	Open answers. The count depends on the coins and the objects.
The glue stick balances 8 coins and the spoon balances 5 coins. Which is heavier?	The glue stick: it needs more of the same coin.
Your partner used erasers and got a different number. Why?	Erasers and coins are different units. To compare, use the same unit.

20–32 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"A watermelon balances 3 pineapples. One pineapple balances a 2 kg weight. The scale is level, so both sides are equally heavy. I can swap each pineapple for 2 kg: 2, 4, 6. The watermelon is 6 kg."*

"Now the watermelon balances one pineapple and 4 bags of sugar. The watermelon is 6 kg and the pineapple is 2 kg, so the bags must be the rest: 6 − 2 = 4 kg. 4 bags make 4 kg, so each bag is 1 kg. I check: 2 + 1 + 1 + 1 + 1 = 6 ✓."

"A jug puzzle: a 4 ℓ jug and a 9 ℓ jug, and I want exactly 1 ℓ. I fill the 9 ℓ jug. I pour into the 4 ℓ jug until it is full: 9 − 4 = 5 left. I empty the 4 ℓ jug and pour again: 5 − 4 = 1. I write every pour in a table so I can check it."

Worked example on the slides	Answer
Balance puzzle: swap it. A watermelon balances 3 pineapples. One pineapple balances a 2 kg weight. The watermelon also balances 1 pineapple and 4 bags of sugar. How heavy is the watermelon? One bag of sugar?	Watermelon 6 kg (2 + 2 + 2). The 4 bags are 6 − 2 = 4 kg, so each bag is 1 kg. Check: 2 + 1 + 1 + 1 + 1 = 6.
Jug puzzle: keep a table. You have a 4 ℓ jug and a 9 ℓ jug with no marks. You can fill a jug, empty it, or pour from one into the other. How can you get exactly 1 ℓ?	Fill the 9 ℓ jug (0, 9). Pour into the 4 ℓ jug (4, 5). Empty the 4 ℓ jug (0, 5). Pour again (4, 1). 1 ℓ is left in the 9 ℓ jug.

SAY *"The key idea today is: same on both sides. A level balance means both sides are equally heavy, so you can swap. For jugs, write every pour in a table. Say it with me."*

DO Show the key idea slide at about minutes 31 to 32. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Balance puzzles: adding the masses on the two sides instead of using "both sides are the same".
- Jug puzzles: writing "pour" without saying how much is in each jug after the pour, so the steps cannot be checked.
- Time: counting on past the hour as if there were 100 minutes in an hour (4:50 + 25 minutes written as 4:75 or 5:75).

32–36 min Brain break

DO Be a balance! Stand on one leg with your arms out. The tutor calls "heavy left!" or "heavy right!" and you tip, then come back level.

Water. Toilet break for anyone who needs it. Keep it to four minutes.

DO Story break (optional, 2 minutes): do the movement game for 2 minutes (32 to 34), then show the "Story break: Thales of Miletus" slide (34 to 36). Read the story aloud (about 30 seconds): "Thales lived in the Greek city of Miletus, which is now in Turkey, about 2,600 years ago. A story told by writers who lived long after him says he measured the height of a pyramid in Egypt. He waited for the time of day when a person's shadow is as long as the person is tall. At that time the pyramid's shadow is as long as the pyramid is tall. (Historians cannot be sure it happened like this.

None of his own writings survive.)" Say clearly: A story told by later writers. Historians cannot be sure it happened like this.

DO Then ask: "Would anyone like to try this one?" Try it: At the moment your shadow is as long as you are tall, a flagpole's shadow is 9 m long. How tall is the flagpole? If your shadow is 120 cm long at that moment, how tall are you? Volunteers only, never name a child, nobody is marked, passing is fine. SKIP the story break, without comment, if you are behind time or the group needs the full four minutes.

36–72 min Worksheet (pairs, then alone, then stretch)

DO Minute 36: give out the workbook. Read the title of Lesson 9 aloud. Say "Pairs first. Talk before you write."

Minutes 36 to 50: pairs work together on the first page (online, visit every shared breakout room by minute 47 so no pair works unseen for long; in Lessons 5, 6 and 10 a short teach slide comes first, 36 to 38). Minutes 50 to 52: check page 1 together. Minutes 52 to 57: short teaching for page 2. Minutes 57 to 69: each child works alone on the second page. Minutes 69 to 72: check page 2 together. Think harder, Go further and the Investigation are open to any child at any time after question 3.

SAY *"Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking."*

DO Circulate and ask:

- Ask: "The scale is level. What does that tell you?"
- Ask: "What can you swap the melon for?"
- Ask: "How much is in each jug now? Show me in your table."

IF STUCK Hint 1: "Find the thing you know first. Which scale has a number on it?"

Hint 2: "Swap: put what balances it on the scale instead."

Hint 3: "For jugs, write the two amounts after every pour."

DO A choice, not a track: after question 3 any child may carry on or try Think harder (the neutral line is printed on page 1). Invite the WHOLE group at once, never one child: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Never announce who is on which page, never name a child in front of the group, and do not split a pair unless both children agree. Children may move either way at any time. The Investigation is printed on the last core page, so every child sees it.

Think harder and Go further questions

Question	Answer
Workbook Think harder 1: Ben cuts a 20 m rope into 4 equal pieces. How long is each piece? How many cuts does he make?	5 m each (5 + 5 + 5 + 5 = 20). 3 cuts: the last piece needs no cut.

Workbook Think harder 2: Mei's family leaves home at 10:45 a.m. They reach Grandma's flat at 12:10 p.m. How long is the trip?	1 hour 25 minutes (10:45 to 11:00 is 15 minutes, 11:00 to 12:00 is 1 hour, 12:00 to 12:10 is 10 minutes).
Workbook Think harder 3: 3 apples balance 1 pomelo. The pomelo and 2 apples together balance a 500 g weight. How heavy is one apple? How heavy is the pomelo?	Apple 100 g, pomelo 300 g (5 apples balance 500 g).
Workbook Think harder 4: You have a 2 ℓ jug and a 7 ℓ jug with no marks. How can you get exactly 3 ℓ? List each pour.	Fill 7, pour into 2, empty 2, pour again: 3 ℓ is left in the 7 ℓ jug.
Workbook Go further 5: You have a 3 ℓ jug and a 7 ℓ jug with no marks. How can you get exactly 5 ℓ? List each pour.	(3 ℓ, 7 ℓ): fill 7 (0, 7); pour (3, 4); empty 3 (0, 4); pour (3, 1); empty 3 (0, 1); pour (1, 0); fill 7 (1, 7); pour until the 3 ℓ jug is full (3, 5). 5 ℓ in the 7 ℓ jug.
Workbook Go further 6: Scale 1: 2 cubes and 1 ball balance 2 balls. Scale 2: 1 ball and 1 cube balance a 300 g weight. How heavy is one cube? How heavy is one ball?	Take one ball off both sides of scale 1: 2 cubes balance 1 ball. On scale 2, swap the ball for 2 cubes: 3 cubes balance 300 g, so a cube is 100 g and a ball is 200 g. Check scale 1: 100 + 100 + 200 = 400 = 200 + 200.
Workbook Go further 7: Mei has 3 lessons in a row. Each lesson is 35 minutes long, with a 10-minute break between lessons. The first lesson starts at 8:15 a.m. What time does the last lesson end?	10:20 a.m. (8:15 to 8:50, break to 9:00, 9:00 to 9:35, break to 9:45, 9:45 to 10:20)

These are the questions on the two Think harder pages in the workbook (Think harder, then Go further). More are in the separate Question Bank.

Investigation (several answers are right)	Model answer and what to look for
Workbook Investigation (last core page): Weights puzzle. You have three different weights. Each weight is 1 g, 2 g, 3 g, 4 g or 5 g. You put some of them on one pan to make a mass. (a) With weights of 1 g, 2 g and 4 g, which masses from 1 g to 7 g can you make? (b) Find ALL the sets of three different weights from 1 g to 5 g that can make every mass from 1 g to 6 g. (c) What do all the good sets have in common? Easy first step: with weights of 1 g, 2 g and 4 g, can you make 3 g? Can you make 5 g? Needs: paper and a pencil (cubes or coins can stand for the weights).	(a) All of them: 1, 2, 1+2 = 3, 4, 1+4 = 5, 2+4 = 6, 1+2+4 = 7. (b) Exactly 2 sets work: 1, 2, 3 (makes 1 to 6) and 1, 2, 4 (makes 1 to 7); brute-forced over all 10 sets. (c) Both contain 1 g and 2 g: only a 1 g weight can make 1 g, and with different weights only a 2 g weight can make 2 g. Look for: a systematic list of the possible sets, checking each mass in order, and the noticing that 1 g is needed.

72–77 min Talk time

DO Bring the group back to a circle (or camera on, mics on). You choose volunteers first: "Who would like to share?" If nobody offers, ask the pairs to share what their partner said. Never cold-

call a child by name. Any child may say "pass" at any time and that is always fine. Do not keep public tallies, scores or rankings (no counting who got most right, no "first to finish"). Invite the whole group to think about whether they would like a turn, never one child.

SAY *"How did swapping help you with the balance puzzles?"*

"Why is a table useful for jug puzzles?"

"How do you count on past the hour?"

Discussion question	Listen for
A melon balances 2 apples. Can you swap the melon on a scale for 2 apples and keep it level? Why?	A level balance means the two sides weigh the same, so the melon can be swapped for 2 apples (or 400 g) and the scale stays level. A table shows how much is in each jug after every pour, so we can check every step and find our way back. Past the hour: count on to the o'clock first (4:50 to 5:00 is 10 minutes), then count the rest.

77–80 min Investigation share (3 minutes)

DO Show the Investigation slide. The Investigation is on the last core page of the workbook and is open to every child from question 3 on. Ask the whole group: "Who would like to share one way they tried it? Passing is fine." Volunteers only, no names, no public tallies. Several answers are right: praise the method, not the speed. If nobody has started, give pairs one minute for the easy first step and set the rest as take-home.

Investigation	Model answer and what to look for
Investigation: Weights puzzle	(a) All of them: 1, 2, 1+2 = 3, 4, 1+4 = 5, 2+4 = 6, 1+2+4 = 7. (b) Exactly 2 sets work: 1, 2, 3 (makes 1 to 6) and 1, 2, 4 (makes 1 to 7); brute-forced over all 10 sets. (c) Both contain 1 g and 2 g: only a 1 g weight can make 1 g, and with different weights only a 2 g weight can make 2 g. Look for: a systematic list of the possible sets, checking each mass in order, and the noticing that 1 g is needed.

80–87 min Exit ticket

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide.

Exit ticket	Answer
A lesson starts at 4:50 p.m. and lasts 25 minutes. What time does it end?	5:15 p.m. (4:50 to 5:00 is 10 minutes, 15 more minutes is 5:15)

87–89 min Looking back (2 minutes)

SAY *"Looking back takes 2 minutes. Find the small box called Looking back on your page. Finish one line, two lines or all three, and tick what helped you. You may pass on any line. This box is for you. If you tick "I am still stuck", your tutor will read it at the next lesson. Nothing is shared with the group."*

DO Nothing is collected in front of the group and nobody has to read theirs out. The Looking back box is for the child. A paper box cannot be read live, so the tutor reads the journals at the next lesson, while stamping, and never by private message or photo. If a child ticked still stuck, start the next open session with a whole-group invitation (for example: "Anyone who would like another look at last lesson's idea, stay with me and we will do it together"), with a parent nearby; never single out one child. A child may volunteer one sentence, never by name unless they choose to. No public tallies. Praise the strategy, not the speed.

89–90 min Closing and stamps

SAY *"You were real Measure Masters today. You swapped things on a balance, you found exact amounts with no marks on the jugs, and you counted time across the hour."*

"Next lesson is Challenge Day: bring your best thinking!"

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide (this is for you only, never shared with the children). If a child marked D or H, plan two minutes of catch-up at the start of the next lesson.

Think harder and Go further

Make your own jug puzzle with two jug sizes and a target amount. Check that it can be done, and list every pour.

Answer key (workbook)

Q1 (a) 200 g (b) 400 g (c) 50 g (the melon is 400 g, the apple is 200 g, so 4 lemons are 200 g) (d) 8 lemons

Q2 (a) Fill the 5 ℓ jug. Pour it into the 3 ℓ jug until the 3 ℓ jug is full. 2 ℓ is left in the 5 ℓ jug. (b) Fill the 3 ℓ jug and pour it into the 5 ℓ jug. Fill the 3 ℓ jug again and pour until the 5 ℓ jug is full (2 ℓ goes in). 1 ℓ is left in the 3 ℓ jug.

Q3 (3 ℓ jug, 5 ℓ jug) Start 0, 0. 1: fill the 5 ℓ jug: 0, 5. 2: pour into the 3 ℓ jug until it is full: 3, 2. 3: empty the 3 ℓ jug: 0, 2. 4: pour the 2 ℓ into the 3 ℓ jug: 2, 0. 5: fill the 5 ℓ jug: 2, 5. 6: pour into the 3 ℓ jug until it is full (1 ℓ fits): 3, 4. 4 ℓ ✓ (6 steps is the shortest way; checked by search.)

Q4 (a) 46 cm (b) 105 cm (64 + 46 = 110, take away the 5 cm overlap)

Q5 (a) A 7:25 a.m. B 4:50 p.m. (b) 5:25 p.m. (c) 45 minutes (10:45 to 11:00 is 15 minutes, 30 more)

Story try-it (Thales of Miletus). The flagpole is 9 m tall. You are 120 cm tall (1 m 20 cm).

Investigation (Weights puzzle). (a) All of them: 1, 2, 1+2 = 3, 4, 1+4 = 5, 2+4 = 6, 1+2+4 = 7. (b) Exactly 2 sets work: 1, 2, 3 (makes 1 to 6) and 1, 2, 4 (makes 1 to 7); brute-forced over all 10 sets. (c) Both contain 1 g and 2 g: only a 1 g weight can make 1 g, and with different weights only a 2 g weight can make 2 g. Look for: a systematic list of the possible sets, checking each mass in order, and the noticing that 1 g is needed.

Lesson 10: Challenge Day

Week 5 (second lesson) • Mixed Puzzles • Workbook: Lesson 10

Objective	Choose a way to start a puzzle (draw it, list it in order, make a table) and use it on mixed puzzles: picture graphs with a key of 5 or 10, a logic grid with exactly one answer, numbers to 1000, money, fractions, and × and ÷ with the 2, 3, 4, 5 and 10 tables. Check the answer and explain it.
Thinking skills	Choosing a strategy, drawing models, listing in order, making tables, logical elimination, checking, explaining
Materials	Four station cards (one slide each in the deck, and printed in the tutor guide), play coins, digit cards 0 to 9, paper strips for fractions, mini whiteboards, certificates, stamps.
Online set-up	Each breakout pair gets one station card (the tutor pastes its text into the breakout room chat), works for 3 minutes, then swaps with another pair for a second card, and everyone reports back to the group. Each child needs digit cards (cut from paper), a pencil and a mini whiteboard.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0–1 min Opening

SAY *"Welcome to Challenge Day, inquirers! Today's puzzles mix everything from our ten lessons."*

"Clever inquirers do not know every answer straight away. They know how to START: draw it, list it, or make a table."

DO Greet each child by name. Show the lesson title slide. Read "Today we will..." aloud.

1–4 min Sneak-peek puzzles (3 minutes)

SAY *"These are sneak-peek puzzles. They are ideas we have not learnt yet, so guessing is welcome and so is saying "I do not know". Nobody is marked and nothing is read out. Start with question 1: use objects, a drawing or counting. Then try question 2. Show your answer on your own paper to your tutor on camera, or keep it to share if you want. If you would like help, the tutor will invite the whole group to ask in the open session, and a parent can stay nearby."*

DO Show the Sneak-peek puzzles slide at minute 1 (3 minutes: about 1 minute for question 1, then 2 minutes for question 2). Say out loud: these are ideas we have not learnt yet; guessing and "I do not know" are welcome; nobody is marked.

Answers stay private. Each child holds their own paper to the camera for you only (look, do not read out), or writes it on their own paper and shows only you on camera. Never raise whiteboards together and never show a board in front of the group. Never ask a named child "how do you know?" Ask for volunteers: "Would anyone like to tell us how they started?"

Question 1 is an access question: every child can try it with objects, a drawing or counting, so everyone has a first win. Question 2 is the thinking question. Do not mark either one. Click to show

the answers at the end as "one way to think about it", without saying who had what.

Use what you see only to plan. Invite the WHOLE group at once, never one child: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Never announce who is on which page and never name a child in front of the group. Children choose, and may move either way at any time. Pairs are not split unless both children agree. Everybody joins the explore, the mini-lesson, the check-together slides and talk time.

Sneak-peek puzzle	Answer and what to listen for
Question 1 (access: objects, drawing or counting): A box has 3 triangles and 2 squares. A triangle has 3 sides and a square has 4 sides. Draw the shapes. How many sides are there altogether?	17 sides (9 + 8). Listen for: Drawing the shapes and counting the sides is a good start. This is a first win for everyone.
Question 2 (thinking): A box has 9 shapes: triangles (3 sides) and squares (4 sides). There are 31 sides altogether. How many squares? Say how you started: draw it, list it or make a table.	4 squares and 5 triangles (16 + 15 = 31 sides). Listen for: Any start is welcome: drawing 9 shapes, listing "1 square, 2 squares ..." or a table. A guess that is checked counts.

DO The neutral line “After question 3 you may choose: carry on, or try Think harder.” is printed on page 1 of the workbook, next to the lesson title. Nobody is labelled and nothing is announced.

4–8 min Warm-up

DO Relay review. Six cards, one from each earlier topic: digit cards, missing digits, legs, patterns, change and fractions. A volunteer answers, then chooses a friend who has not had a turn (pass is always fine).

SAY *“Relay time! I show a card and ask for a volunteer. That person answers, then chooses a friend who has not had a turn. You can always say pass. A different answer gets “let us test that”, never a groan.”*

DO Keep the game to two minutes (4 to 6). Then the quick questions (6 to 8): ask questions 1, 3 and 5; questions 2 and 4 are spare. Children answer by thumbs, a whisper to a partner. Do not name a child.

Ask	Expect
Share 40 stickers equally among 5 children. How many each?	8
It is 10:55 a.m. What time is it 10 minutes later?	11:05 a.m.
Which is bigger: 1/3 or 1/5 of the same cake?	1/3
Half of a circle is called a ...	Semicircle
I think of a number and add 30. I get 100. What is my number?	70

8–20 min Explore (in pairs, with objects)

SAY *"Look at the four station cards. Which one will be hardest? Which way will you start it: draw, list or table? Write your guess."*

DO 1. Your pair gets one station card: a picture graph, a logic grid, a coin puzzle or a fraction strip.

2. Read the card aloud. Before you solve it, write how you will start: draw, list or table.

3. Solve it and check it (3 minutes). Then swap cards with another pair.

4. Be ready to explain how you started and how you know.

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK Stuck: "Can you draw it? Can you list the ways in order?"

Graph card: "Look at the key first. What is one picture worth?"

Finished early: "Is there another answer? How do you know there is not?"

DO Station cards (each is also one slide in the deck). One card per breakout pair, about 3 minutes, then swap:

Station card	Answer
Card 1: picture graph. Each picture stands for 5 books. Monday has 3 pictures, Tuesday 5 pictures, Wednesday 2 pictures. How many books were read on these 3 days?	50 books (10 pictures × 5)
Card 2: logic grid. Ali, Ben and Cai each have one pet: a cat, a dog or a fish. Ben has the fish. Ali does not have the dog. Who has the cat?	Ali has the cat (Ben has the fish, so Ali and Cai share the cat and the dog; Ali does not have the dog, so Ali has the cat and Cai has the dog)
Card 3: coin puzzle. Make 70¢ with exactly 4 coins. You can use 5¢, 10¢, 20¢ and 50¢ coins. Find all the ways.	2 ways: 50¢ + 10¢ + 5¢ + 5¢, and 20¢ + 20¢ + 20¢ + 10¢
Card 4: fraction strip. A strip is cut into 8 equal parts. 3 parts are red and 2 parts are blue. What fraction is NOT coloured?	3/8

DO Go further (a second what-if, so nobody waits): show the Explore: go further slide or post it in the chat when pairs go to their rooms. Any child may try it any time, as a choice. Never name who does. Question: what if the coin puzzle is 70¢ with exactly 5 coins (using 5¢, 10¢, 20¢ and 50¢ coins)? Can it be done? How? Answer: Yes: 50¢ + 5¢ + 5¢ + 5¢ + 5¢; 20¢ + 20¢ + 20¢ + 5¢ + 5¢; 20¢ + 20¢ + 10¢ + 10¢ + 10¢.

DO At about minute 14, say "Pencils down" and bring the group back together. Take the three questions below, one at a time (about 6 minutes).

Ask	Expect

How did you start each card?	Open answers. Draw it, list it, or make a table.
Which card was hardest? Why?	Open answers.
What helped when you were stuck?	Drawing it, going in order, trying small numbers first, checking with the clues.

20–32 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"Mei had some stickers. She gave half to Ben. Then she gave 6 to Siti and had 4 left. I cannot hold that in my head, so I draw a bar cut into two halves. One half went to Ben. The other half is the 6 for Siti and the 4 left: 6 + 4 = 10. Both halves are the same, so Mei had 10 + 10 = 20 stickers. Check: half of 20 is 10, 10 – 6 = 4 ✓."*

"Digit cards 2, 0 and 5. How many 3-digit numbers? I list them in order, smallest hundreds digit first. 0 cannot go first. 2: 205, 250. 5: 502, 520. Four numbers, and I know I have them all because I went in order."

"Chickens and goats: 7 heads and 20 legs. I make a table. 0 goats: 14 legs. 1 goat: 16. 2 goats: 18. 3 goats: 20 ✓. So 3 goats and 4 chickens. Each goat I swap in adds 2 legs."

Worked example on the slides	Answer
Draw it: a before-and-after story. Mei had some stickers. She gave half of them to Ben. Then she gave 6 to Siti. Now she has 4 left. How many stickers did she have at first?	20. The half she kept is 6 + 4 = 10, so she had 10 + 10 = 20. Check: 20 → 10 → 4.
Make a table: heads and legs. A farm has chickens and goats. There are 7 heads and 20 legs. How many goats are there?	3 goats and 4 chickens. Table: 0 goats 14 legs, 1 goat 16, 2 goats 18, 3 goats 20 ✓. Check: 12 + 8 = 20.

SAY *"The key idea today is: how to start a puzzle. Stuck? Draw it, list it in order, or make a table. Say it with me."*

DO Show the key idea slide at about minutes 31 to 32. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Picture graph: counting the pictures instead of using the key (writing 6 instead of 30 for Tuesday).
- Listing: putting 0 first (038 is not a 3-digit number) or missing a number because the list is not in order.
- Logic grid: putting a tick without crossing out the rest of that row and column, so the next clue cannot be used.

32–36 min Brain break

DO Freeze with maths. Dance until the music stops, then make the shape the tutor calls with your arms: a semicircle, a quarter circle, a cone, a cuboid.

Water. Toilet break for anyone who needs it. Keep it to four minutes.

DO Story break (optional, 2 minutes): do the movement game for 2 minutes (32 to 34), then show the "Story break: Paul Erdos" slide (34 to 36). Read the story aloud (about 30 seconds): "Paul Erdos was born in Budapest, Hungary, in 1913. He loved maths problems, and he loved working on them with other people. He travelled from city to city, carrying very little, and worked with more than 500 different mathematicians. Together they wrote well over a thousand papers. He also set hard problems and offered cash prizes to anyone who solved them. For Erdos, a good problem was something to share."

DO Then ask: "Would anyone like to try this one?" Try it: Solve this with a partner. Two friends have 12 sweets between them. One has 4 more than the other. How many sweets does each friend have? Volunteers only, never name a child, nobody is marked, passing is fine. SKIP the story break, without comment, if you are behind time or the group needs the full four minutes.

36–72 min Worksheet (pairs, then alone, then stretch)

DO Minute 36: give out the workbook. Read the title of Lesson 10 aloud. Say "Pairs first. Talk before you write."

Minutes 36 to 50: pairs work together on the first page (online, visit every shared breakout room by minute 47 so no pair works unseen for long; in Lessons 5, 6 and 10 a short teach slide comes first, 36 to 38). Minutes 50 to 52: check page 1 together. Minutes 52 to 57: short teaching for page 2. Minutes 57 to 69: each child works alone on the second page. Minutes 69 to 72: check page 2 together. Think harder, Go further and the Investigation are open to any child at any time after question 3.

SAY *"Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking."*

DO Circulate and ask:

- Ask: "How did you start this one? Draw, list or table?"
- Ask: "How do you know there is only one answer?"
- Ask: "Show me your check."

IF STUCK Hint 1: "Which lesson does this remind you of?"

Hint 2: "Draw it, list it in order, or make a table."

Hint 3: "Try a small number first, then change it one step at a time."

DO A choice, not a track: after question 3 any child may carry on or try Think harder (the neutral line is printed on page 1). Invite the WHOLE group at once, never one child: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Never announce who is on which page, never name a child in front of the group, and do not split a pair unless both children agree. Children may move either way at any time. The Investigation is printed on the last core page, so every child sees it.

Think harder and Go further questions

Question	Answer

Workbook Think harder 1: 30 children chose a fruit. In the picture graph each picture stands for 5 children. Apple has 2 pictures and banana has 3 pictures. The mango row is missing. How many pictures should the mango row have?	1 picture (30 − 10 − 15 = 5 children)
Workbook Think harder 2: How many 3-digit numbers have digits that add up to 3? List them all.	6: 102, 111, 120, 201, 210, 300
Workbook Think harder 3: A bus had some people on it. At the first stop half of the people got off. At the second stop 6 people got on. Now there are 14 people on the bus. How many people were on the bus at the start? Draw a bar to start.	16 (14 − 6 = 8 is half, 8 + 8 = 16).
Workbook Think harder 4: 5, 9, 13, 17, … What is the 10th number in this pattern?	41 (add 4: 21, 25, 29, 33, 37, 41)
Workbook Go further 5: A picture graph has lost its key. Football has 6 pictures and swimming has 4 pictures. 10 more children chose football than swimming. How many children chose swimming? Football?	Swimming 20, football 30 (2 pictures stand for 10 children, so 1 picture stands for 5)
Workbook Go further 6: Ali, Ben, Mei and Siti each have one coin: 10¢, 20¢, 50¢ or $1. Mei has more money than Ali but less than Ben. Ben does not have the $1 coin. Who has which coin?	Ali 10¢, Mei 20¢, Ben 50¢, Siti $1. Ben is not $1 and two people have less than him, so Ben has 50¢; then Ali 10¢ and Mei 20¢, and Siti has the $1.
Workbook Go further 7: At a hawker centre, chicken rice costs $4 a plate and noodles cost $3 a bowl. Ravi's family buys 7 dishes for $25. How many plates of chicken rice do they buy? What is the smallest total 7 dishes can cost? The biggest? Can 7 dishes cost $30? Can they cost EVERY whole-dollar amount in between?	4 plates of chicken rice and 3 bowls of noodles ($16 + $9 = $25). Table: 0 plates → $21, 1 → $22, 2 → $23, 3 → $24, 4 → $25 ✓. Smallest $21 (all noodles), biggest $28 (all chicken rice), so $30 is impossible. Each swap of a bowl for a plate adds exactly $1, so every whole dollar from $21 to $28 is possible.

These are the questions on the two Think harder pages in the workbook (Think harder, then Go further). More are in the separate Question Bank.

Investigation (several answers are right)	Model answer and what to look for

Workbook Investigation (last core page): Staircase. Mei climbs a staircase. She can take 1 step or 2 steps at a time. For 3 steps there are 3 ways: 1+1+1, 1+2 and 2+1. (a) Find ALL the ways for 4 steps. (b) Find ALL the ways for 5 steps. (c) Fill in a table for 1, 2, 3, 4 and 5 steps. What pattern do you see? Can you guess the number of ways for 6 steps? Easy first step: draw 3 steps and show the 3 ways: 1+1+1, 1+2 and 2+1. Needs: paper and a pencil.	Number of ways: 1 step: 1, 2 steps: 2, 3 steps: 3, 4 steps: 5, 5 steps: 8 (brute-forced; 6 steps: 13). 4 steps: 1+1+1+1, 1+1+2, 1+2+1, 2+1+1, 2+2. Pattern: each number is the sum of the two before it (3 + 5 = 8, 5 + 8 = 13). Look for: a list in order (all 1s first, then swap in a 2), a table, and the pattern spotted from the table. Do not name Fibonacci numbers unless a child asks.

72–77 min Talk time

DO Bring the group back to a circle (or camera on, mics on). You choose volunteers first: "Who would like to share?" If nobody offers, ask the pairs to share what their partner said. Never cold-call a child by name. Any child may say "pass" at any time and that is always fine. Do not keep public tallies, scores or rankings (no counting who got most right, no "first to finish"). Invite the whole group to think about whether they would like a turn, never one child.

SAY *"Which puzzle was hardest today? How did you start it?"*

"When is a table better than a drawing?"

Discussion question	Listen for
Pick one puzzle and teach it to the group in two sentences: how you started it, and how you know your answer is right.	Open answers. Listen for a clear start (I drew bars, I listed in order, I made a table) and a check (I put my answer back into the puzzle). A table is good when we try 0, 1, 2, 3 … and something grows each time; a drawing is good for before-and-after stories and more-than or fewer-than.

77–80 min Investigation share (3 minutes)

DO Show the Investigation slide. The Investigation is on the last core page of the workbook and is open to every child from question 3 on. Ask the whole group: "Who would like to share one way they tried it? Passing is fine." Volunteers only, no names, no public tallies. Several answers are right: praise the method, not the speed. If nobody has started, give pairs one minute for the easy first step and set the rest as take-home.

Investigation	Model answer and what to look for
Investigation: Staircase	Number of ways: 1 step: 1, 2 steps: 2, 3 steps: 3, 4 steps: 5, 5 steps: 8 (brute-forced; 6 steps: 13). 4 steps: 1+1+1+1, 1+1+2, 1+2+1, 2+1+1, 2+2. Pattern: each number is the sum of the two before it (3 + 5 = 8, 5 + 8 = 13). Look for: a list in order (all 1s first, then swap in a 2), a table, and the pattern spotted from the table. Do not name Fibonacci numbers unless a child asks.

80–85 min Exit ticket

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide.

Exit ticket	Answer
5 bicycles and tricycles have 13 wheels altogether. How many tricycles are there? How did you start?	3 tricycles and 2 bicycles (9 + 4 = 13). A table of 0, 1, 2, 3 tricycles works well.

85–87 min Looking back (2 minutes)

SAY *"Looking back takes 2 minutes. Find the small box called Looking back on your page. Finish one line, two lines or all three, and tick what helped you. You may pass on any line. This box is for you. If you tick "I am still stuck", your tutor will read it at the next lesson. Nothing is shared with the group."*

DO Nothing is collected in front of the group and nobody has to read theirs out. The Looking back box is for the child. A paper box cannot be read live, so the tutor reads the journals at the next lesson, while stamping, and never by private message or photo. If a child ticked still stuck, start the next open session with a whole-group invitation (for example: "Anyone who would like another look at last lesson's idea, stay with me and we will do it together"), with a parent nearby; never single out one child. A child may volunteer one sentence, never by name unless they choose to. No public tallies. Praise the strategy, not the speed.

87–90 min Certificates and closing

SAY *"You are real inquirers now: you notice it, test it and explain why. And when you are stuck, you know how to start."*

"Give each child a certificate and say one specific thing they did well in the program."

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide (this is for you only, never shared with the children). If a child marked D or H, plan two minutes of catch-up at the start of the next lesson.

Think harder and Go further

Write a challenge puzzle for next year's group, with the answer and a hint that says how to start it. Check it has exactly one answer.

Answer key (workbook)

Q1 (a) 30 (b) Thursday sold 25 ice-creams: draw 5 pictures (90 − 20 − 30 − 15 = 25) (c) $30 (15 ice-creams at $2 each)

Q2 (a) 308, 380, 803, 830 (b) Greatest 830, smallest 308 (c) 308, 380, 803

Q3 Two 50¢ coins and four 20¢ coins (100¢ + 80¢ = 180¢ = $1.80). Table: 0 → $1.20, 1 → $1.50, 2 → $1.80 ✓

Q4 Ali: car. Ben: MRT. Mei: bus. Siti: walk.

Q5 (a) 5/10 (b) 2/10 (c) Mei, Ali, Ben

Q6 (a) 16, 20, 24 cubes in Towers 4, 5, 6; Tower 8 has 32 cubes (each tower adds 4: 4, 8, 12, 16, 20, 24, 28, 32) (b) Tower 9 (36 cubes: 4 × 9, or keep adding 4: 32, 36)

Story try-it (Paul Erdos). 8 and 4. Check: 8 + 4 = 12 and 8 − 4 = 4.

Investigation (Staircase). Number of ways: 1 step: 1, 2 steps: 2, 3 steps: 3, 4 steps: 5, 5 steps: 8 (brute-forced; 6 steps: 13). 4 steps: 1+1+1+1, 1+1+2, 1+2+1, 2+1+1, 2+2. Pattern: each number is the sum of the two before it (3 + 5 = 8, 5 + 8 = 13). Look for: a list in order (all 1s first, then swap in a 2), a table, and the pattern spotted from the table. Do not name Fibonacci numbers unless a child asks.

Progress tracker

After each exit ticket, write S (secure: correct and can explain), D (developing: correct but cannot explain yet) or H (needs help). Use the notes for what to revisit next lesson.

Student	L1	L2	L3	L4	L5	L6	L7	L8	L9	L10	Notes

Ending the program

- In Lesson 10, give each child a certificate and their completed inquiry journal.
- Send parents a short message through the approved channel: one thing their child did well, and one thing to keep practising.
- Delete or hand over all Student Data within 7 days after the last lesson (this is in the tutor agreement).

Tutor checklist

Before each lesson

- Read the lesson plan and solve the worksheet yourself without looking at the key.
- Check materials are ready, or the online set-up has been sent.
- Test the platform, screen share and annotation tools 10 minutes early (online).
- Print or open the slide deck for the lesson.

During the lesson

- Circulate and ask questions rather than giving answers.
- Use only the approved channels to contact parents. No private messages to children. Invite the whole group, never one child. If a child ticks “I am still stuck” on the paper Looking back box, read it at the next lesson and offer a whole-group follow-up in the open session, with a parent nearby. Never use personal accounts.
- Do not take photos or recordings unless the Company has approved it and parents have consented.
- If you have a concern about a child’s safety or welfare, tell the Company straight away.

After the lesson

- Stamp inquiry journals and fill in the tracker.
- Collect worksheets and answer keys (face to face) and keep them secure.
- Note any child who needs help, and plan a short catch-up at the start of the next lesson.

Math Inquirers Holiday Program

Notice it. Test it. Explain why.

A 10-lesson holiday maths program for children whose school year covers Primary 2 topics

Your child will spend the holidays as a young mathematician: noticing patterns, testing ideas with real objects and explaining their thinking. Lessons are live online in small groups, hands-on, and built around thinking, not speed.

What your child will do

- Lesson 1, Digit Detectives: numbers to 1000
- Lesson 2, Missing Digit Mysteries: addition and subtraction within 1000
- Lesson 3, Legs, Wheels and Arrays: multiplication and division
- Lesson 4, Pattern Machines: patterns with multiplication
- Lesson 5, Mystery Number Machines: working backwards
- Lesson 6, Market Day: money
- Lesson 7, Fair Shares: fractions
- Lesson 8, Shape Builders: 2d and 3d shapes
- Lesson 9, Measure Masters: length, mass, volume and time
- Lesson 10, Challenge Day: mixed puzzles

How you can help

- Ask “How do you know?” when your child tells you an answer.
- Let your child try first. Please sit nearby, but let them find the answer.
- Print the workbook and cut-outs, and keep counters, scissors and a pencil ready.

Practical details

Dates	[start date] to [end date], two lessons a week
Time	[day] and [day], [start time] to [end time]
Where	Live online video class (link sent after registration)
Group size	[number] children
Fee	S$500 introductory (regular S$650), workbook included, payable in 3 instalments
Contact	[name, phone or email]

Math Inquirers Holiday Program is an independent enrichment program. It is inspired by enrichment-style thinking skills but is not affiliated with, run by or endorsed by the Ministry of Education or any school.